

Low-Speed Aerodynamic Characteristics of a Delta Wing with Deflected Wing Tips

Thesis

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By

Colin Weidner Trussa

Graduate Program in Aeronautical and Astronautical Engineering

The Ohio State University

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Master's Examination Committee

Dr. Clifford Whitfield, Advisor

Dr. Rick Freuler

Dr. Matthew McCrink

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ABSTRACT

The purpose of this work was to investigate the low-speed aerodynamic characteristics of a novel delta wing layout with deflected wing tips. This project is motivated by the ongoing unmanned aerial vehicle research and development at The Ohio State University Aerospace Research Center. The model under test for this study had four main design requirements: (1) high-speed, (2) highly maneuverable, (3) aerodynamically interesting, and (4) multi-configurable. The last three requirements are addressed directly in this report with specific emphasis on requirements two and three. A modular fuselage design satisfied requirement four, and the novel delta wing addressed requirements two and three.

The novel delta wing has a leading-edge sweep of 60 degrees, a high-speed airfoil with a rounded leading-edge, and wing tips that can rotate a full 180 degrees about a hinge, located at $2/3^{\text{rds}}$ of the half-span parallel to fuselage centerline. Three different wing tip deflection configurations were analyzed: positive, negative, and asymmetric. Positive wing tip deflection corresponds to the wing tips being deflected up towards the vertical tail. Negative wing tip deflection is when the wing tips are deflected down, away from the vertical tail. While the asymmetric configuration has one wing tip deflected up and the other down.

Analysis on the model was completed using a panel method code and experimental wind tunnel testing. The panel method code used was OpenVSP. Upon implementing a vortex lift factor, it was determined that the delta wing results from OpenVSP were only useful for lift related data after comparing the panel method results to theory and publicly available delta wing data. The wind tunnel used for this work is located at the Aerospace Research Center. The wind tunnel is an open circuit subsonic wind tunnel with a 3'x5' test section. Aerodynamic forces were measured using an internal six-component force balance. Tests were performed at two different Reynolds numbers, 3.65×10^5 and 5×10^5 . Significant results from wind tunnel testing found that as the wing tip deflection is increased, for all three layouts, the static longitudinal stability of the model decreased with no significant loss in lift characteristics. The adjustable stability as a result of rotating a delta wing's wing tips, provides inflight adjustments to the maneuverability characteristics of the aircraft.

DEDICATION

Dedicated to my mom.

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VITA

May 13, 1994 Born, Middleburg Heights, USA

May 6, 2018 B.S. Aeronautical and Astronautical
Engineering, The Ohio State University

August 21, 2018 – Present Graduate Research/Teaching Associate,
Aeronautical and Astronautical Engineering,
The Ohio State University

Fields of Study

Major Field: Aeronautical and Astronautical Engineering

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CHAPTER 1. INTRODUCTION

The Ohio State University Aerospace Research Center (ARC) has ongoing tests into autonomous unmanned aerial vehicles (UAV). At the ARC there have been significant advancements in autonomous UAV flight capability and in-flight data acquisition while using commercially available small aircraft [1]. In order to continue to advance research and development on multiple fronts, including flight testing data acquisition development, a new unique aircraft design was required. The requirements for the new design are provided in Table 1.

Table 1. Aircraft Design Requirements

Requirement	Definition
High-speed	Aerodynamically efficient with airframe and engine growth
Highly maneuverable	Inherently capable for low-speed high g-load maneuvers
Aerodynamically interesting	Nonlinear aerodynamic characteristics due to complex flows
Multi-configurable	Easily adaptable for multiple wing configurations

Upon considering the design requirements, a baseline aircraft design was developed by the Flight Vehicle Design and Testing Group at the ARC. The baseline aircraft consisted of an axisymmetric fuselage, a single vertical tail and a delta wing. The delta wing has a

60-degree leading-edge wing sweep and a high-speed symmetric airfoil with a rounded leading-edge. The aircraft configuration was designed using area-ruling techniques in order to ultimately reduce high-speed compressible wave-drag effects due the aircraft's profile.

The baseline wing design delivered a platform that achieved all four major design requirements. The 60-degree swept wing provided benefits for high-speed flight, while the inherent nonlinear lift characteristics at low-speed flight provided the desired complex flows and potentially increased maneuverability characteristics. The overall delta wing layout provided an intrinsically robust structural design for high g-load maneuvers while also providing a simple platform for the ability to easily incorporate additional wing design features. Finally, variations of the 60-degree delta wing configuration and corresponding performance characteristics in low-speed flight had been broadly studied, providing an established baseline for future research and development efforts. The focus of this discussion will be on the first multi-configurable platform that consisted of the 60-degree wing design with articulating wing tips.

The desire to advance the development of highly maneuverable UAVs in-flight data acquisition systems and flight capabilities also extends to recent U.S military needs. As stated by the United States Defense Science Board Task Force on Unmanned Aerial Vehicles and Uninhabited Combat Aerial Vehicles, [2], there is a clear need for an increase in UAVs in the military combat zone. With the demand for more versatile unmanned aircraft capable of operating in high threat areas, there is an inherent need for the UAV to have increased survivability. One way this can be achieved is through increased maneuverability of the UAVs.

As defined by Herbst [3], there are two combat zones for a maneuverable aircraft, short range and medium range. Short-range combat pushes the aircraft through changes in total energy due to hard maneuvers while flying at a low and constant speed. Medium-range combat requires careful total energy management by the pilot while flying supersonic. The aircraft configuration that will be discussed here falls under the short-range combat zone.

1.1 Subsonic Aerodynamic Characteristics of Delta Wings

The flow surrounding a delta wing in low-speed, high angle-of-attack flight inherently leads towards nonlinear aerodynamic characteristics. The nonlinear aerodynamic characteristics are due to very complex flow features that are dependent on geometry and flight conditions; and in the context of this research, there are four features of interest: (1) the leading-edge vortex, (2) how the leading-edge shape impacts delta wing performance, (3) the effect of Reynolds and Mach Number on the leading-edge vortex, and (4) various passive and active methods used to control the leading-edge vortex. This section will introduce and discuss each feature-of-interest, identifying its importance and relevance for the design study, which led to the motivation and objectives of the research.

1.1.1 Leading-Edge Vortex

A key flow feature of a delta wing is the leading-edge vortex that forms at high angles-of-attack in low-speed flight. The importance of the leading-edge vortex is due to

the added lift contribution that is a result of the vortex. This extra vortex lift component drastically increases the stall angle-of-attack compared to that of a more conventional low leading-edge swept wing. Gursul et al. has collected an extensive review of historical and current unsteady aerodynamic characteristics of delta wings with leading-edge sweeps less than or equal to 55 degrees [4]. Delta wings with a sweep less than 60 degrees are considered non-slender delta wings, while delta wings with sweeps greater than or equal to 60 degrees are considered slender. They have explained that the primary leading-edge vortex forms due to boundary layer separation at the leading-edge that rolls up into a primary vortex. More specifically, the leading-edge vortex is the combination of discrete vortices shedding off the leading-edge that form a primary vortex [5]. Once the primary vortex is fully formed, it interacts with the boundary layer on the surface of the wing and promotes further boundary layer separation that results in a secondary counter rotating vortex. An example of the vortex structures developed on a delta wing can be seen in Figure 1.

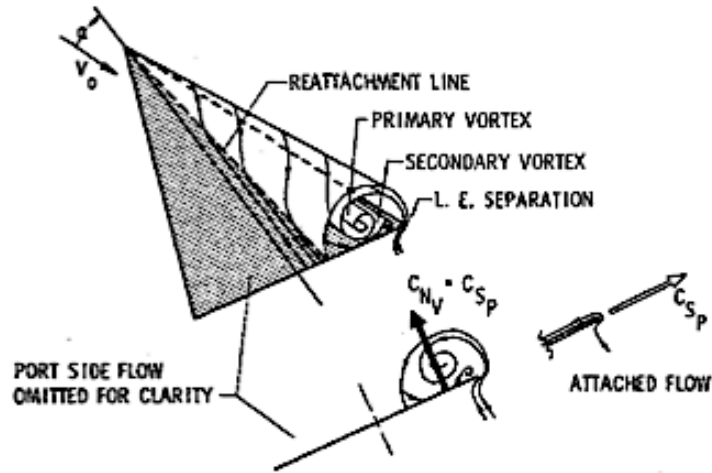


Figure 1. Vortex Flow Illustration [6]

The mechanism for the formation of the secondary vortex has been found to be caused by a large lateral pressure gradient at the inboard reattachment location of the primary vortex. The boundary layer on the surface of the wing at the inboard primary vortex reattachment line then separates and rolls up into the secondary counter rotating vortex [7]. The nature of the boundary layer on the wing's surface, either laminar or turbulent, was also found to greatly impact the formation of the secondary vortex. When there is a laminar boundary layer the secondary vortex is much larger than when a secondary vortex develops from a turbulent boundary layer [7].

Lift of a delta wing at low-speeds can be separated into two components: vortex lift and potential lift. The onset of vortex lift is known as the leading-edge suction analogy, defined by Polhamus [6, 8]. The leading-edge suction analogy holds for the case when the

vortex induces reattached flow, resulting in a vortex and potential flow component of the lift and drag equation, Equation 2 and 3.

$$C_L = K_p \sin \alpha \cos^2 \alpha + K_v \sin^2 \alpha \cos \alpha \quad (2)$$

$$C_D = C_{D_0} + K_p \sin^2 \alpha \cos \alpha + K_v \sin^2 \alpha \quad (3)$$

The K_p and K_v terms correspond to the potential and vortex lift coefficients respectively. The leading-edge suction analogy can be applied to thin wings with no camber or twist. The relationship between the wing's aspect ratio and the values of the two coefficients can be seen below in Figure 2.

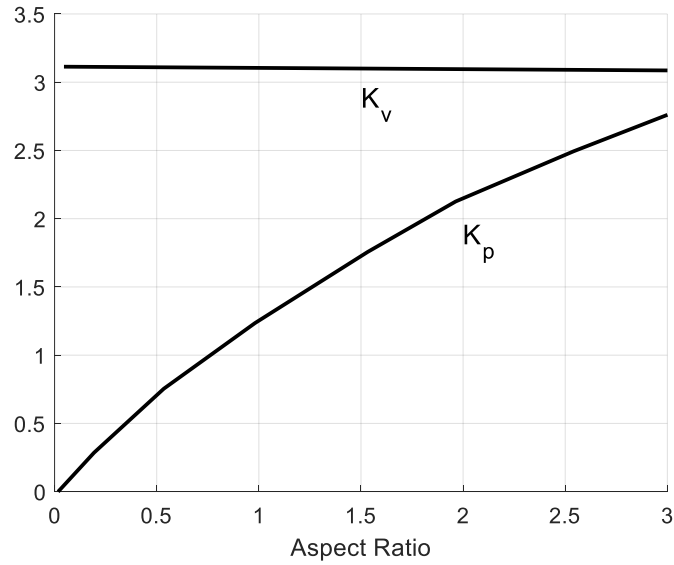


Figure 2. Potential and Vortex Lift Coefficients with Aspect Ratio [6]

By understanding how the geometry of a delta wing impacts the strength of each parameter in the leading-edge suction analogy, a designer can make informed decisions of how a delta wing platform will behave aerodynamically. These detailed aerodynamic and geometric trends are very useful prior to detailed design and wind tunnel testing and extensive computational analyses.

To further understand the leading-edge vortex element, it is important to identify the vortex breakdown, or commonly referred to as vortex “bursting.” A leading-edge vortex breakdown occurs at high angles-of-attack, when the axial flow of the vortex decreases rapidly, which in turn causes the vortex to grow and eventually decay into turbulence. There are two common types of vortex breakdown, bubble and spiral [9]. A bubble vortex breakdown is defined as a stagnation point in the axis of the vortex that causes an oval shaped recirculation bubble and then results in turbulent flow, shown in Figure 3 a). A spiral type breakdown occurs when the vortex core flow experiences a rapid deceleration and develops a spiral flow pattern before also breaking apart into turbulent flow, shown in Figure 3 b).

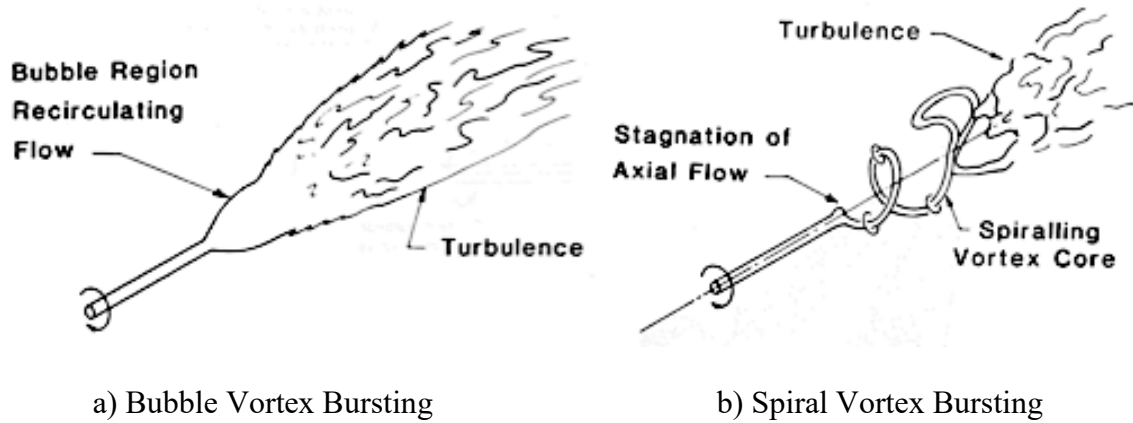


Figure 3. Common Types of Vortex Bursting [9]

Another important flow characteristic to understand delta wing aerodynamic instability characteristics is the roll inducing vortex bursting, known as wing rock. Wing rock is defined as “a high angle-of-attack dynamic phenomenon of limit cycle motion predominantly in roll” [10]. One cause of wing rock is attributed to nonlinear aerodynamic characteristics of the static rolling and yawing moments, while at high angles-of-attack. Wing rock vortex bursting at high angles-of-attack near stall has been documented for delta wings similar to the one in this study [11].

1.1.2 Leading-Edge Geometry

Leading-edge geometry directly impacts the behavior of vortex separation. Mat et al. has identified on a 65-degree swept delta, that as the leading-edge radius is increased the primary vortex structure decreases, the size of the inboard vortex structure increases,

and the inboard vortex separation line moves towards the leading-edge [12]. A flow topology illustration, is provided below in Figure 4, which identifies the vortex terminology and corresponding locations along the wing [13].

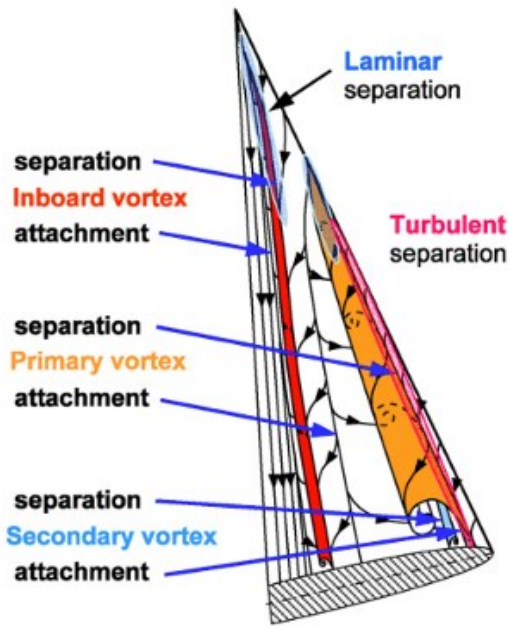


Figure 4. Flow Topology of a Rounded Leading-Edge Delta Wing [13]

Another distinct flow characteristic of a rounded leading-edge delta, is that in between the leading-edge and primary vortex separation line there is attached flow. The vortex sensitivity to leading-edge shape was determined by Luckring [14] for a 65-degree swept delta. Luckring found that a vortex from a sharp leading-edge tends to separate

further upstream on the leading-edge, while a vortex from a blunt leading-edge separates further downstream on the leading-edge. While understanding the behavior of the flow on a delta wing with various leading-edges is necessary, it is equally important to know how the leading-edge geometry influences aerodynamic performance.

Gursul et al. identified that for a 50-degree swept delta wing the effect of leading-edge shape only began to impact the lift curve slope near the stall condition [4]. Up until angles-of-attack of 20 degrees, the lift curve slopes of both a sharp and rounded leading edge were consistent. After 20 degrees of pitch, the non-slender delta wing with a rounded leading edge had a larger maximum lift coefficient than that of a sharp leading-edge. The higher maximum lift values result is a less severe stall condition for the rounded leading-edge delta. Conversely, for slender delta wings, leading-edge geometry was not found to significantly impact aerodynamic performance, [15].

1.1.3 Reynolds and Mach Number Effects on the Leading-Edge Vortex

The leading-edge vortex has also been found to be sensitive to Mach and Reynolds number, specifically Luckring identified the relationship between separation location of the leading-edge vortex with changes for both parameters. With Mach number held constant ($M = 0.4$) and Reynolds number increased (6 – 60 million), the leading-edge vortex separation is delayed further downstream of the leading-edge. On the other hand, as Mach number is increased ($M = 0.4 - 0.6$) and Reynolds number held constant (60 million), the leading-edge vortex separation point is more upstream on the leading-edge [14]. Mat et al. has also identified that as Reynolds number increases, the primary leading-edge

vortex forms further downstream on the wing and the inboard vortex shifts towards the leading edge, [12]. The impact on the lift coefficient and potential lift theory with increasing Reynolds numbers was identified by Henderson [16]. It was found that as Reynolds number increases, the angle-of-attack at which the experimental lift curve deviates from potential flow lift increases.

1.1.4 Methods Used to Control the Leading-Edge Vortex

Due to the significance of the leading-edge vortex, active methods of controlling the vortex have been explored. Recently, techniques using active flow control devices have also been studied; for example, Gad-el-Hak [17] performed a series of tests with ports at the leading-edge of the delta that would periodically eject high pressure air in line with the leading-edge. It was found that the leading-edge vortex could be changed using various pulsed air frequencies from the leading-edge ports.

A method of controlling the leading-edge vortex that is more in-line with this research is passive vortex control. A common passive control method is the leading-edge vortex flap. The leading-edge vortex flap can be a significant addition to the aircraft when trying to improve maneuvering capabilities, through impacting the wing's lift-to-drag ratio. Rinoie [18] performed a series of tests on leading-edge vortex flaps of a 60-degree delta that assessed how the leading-edge radius impacts the flap's performance. It was found that by increasing the flaps leading-edge radius, drag is drastically reduced. However, this decrease in drag was not observed in the minimum drag region with lift coefficients ranging from 0.0 - 0.25. Rinoie also discovered that the deflected rounded leading-edge vortex flap,

when compared to the deflected sharp flap, had higher lift-to-drag ratios for large lift coefficients [18]. The largest improvements were seen between lift coefficients of 0.6 and 0.8. While the lift-to-drag ratio did improve over a range when using a rounded leading-edge flap, the absolute maximum lift-to-drag ratio did not vary with the sharp leading-edge flap. Marchman also has identified the benefits of a leading-edge vortex flap with reductions in drag and improvements in lift-to-drag ratio for a 60-degree delta [19]. Interestingly, Marchman found that the leading-edge vortex flap could alter the longitudinal characteristics of a delta wing, Figure 5.

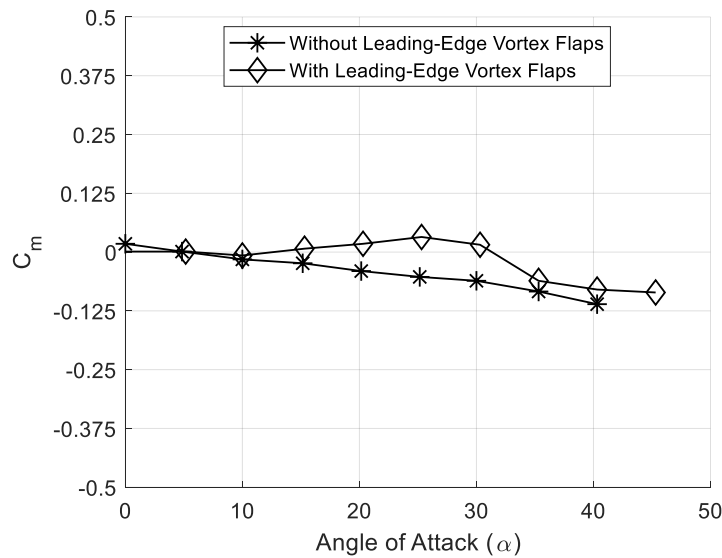


Figure 5. Effect of Leading-Edge Vortex Flaps on Longitudinal Stability [19]

Marchman also studied how an inverted leading-edge vortex flap would impact the aerodynamic characteristics of a delta wing [19]. A typical leading-edge flap is deflected down, whereas an inverted is deflected upwards (in the aircraft reference frame). It was discovered that an inverted flap greatly improved lift for low angles-of-attack, with a 60% increase in lift coefficient at an angle-of-attack of 5 degrees for the delta wing studied.

1.2 Motivation

The wing design selected to satisfy the high maneuverability and nonlinear lift characteristics was a delta wing with deflected wing tips. Deflected wing tips have been incorporated on delta wings for two specific reasons: increased lift due to shock capture at supersonic speeds, and improved stability at supersonic speeds. There are two specific issues regarding stability at supersonic speeds; at supercruise the effectiveness of the vertical tail is diminished, which reduces directional stability, and the aerodynamic center shifts downstream, causing additional longitudinal stability complications [20]. By rotating the wing tips down at supersonic speeds there is a larger effective vertical tail area (increases directional stability) and the aerodynamic center does not shift downstream, eliminating inherent longitudinal stability changes. The advantages of incorporating rotating wing tips in supersonic speeds led to the motivation of determining if performance advantages could also be utilized during subsonic flight by fully articulating the wing tip deflection angles.

1.3 Objectives

The main objective of this research was to determine the effect of deflected wing tips on low-subsonic flight maneuverability and stability through simulation and scaled wind tunnel testing. For exploring current simulation capabilities, a panel method code, OpenVSP, was used to identify if it can accurately predict the aerodynamic characteristics of a standard delta wing. After comparing the accuracy of OpenVSP's data to theory and known results for a standard delta wing, wind tunnel testing was conducted to assess the aerodynamic characteristics of the novel deflected wing tip design. Wind tunnel tests were conducted at ARC's 3' x 5' low-speed, Eiffel-type open circuit subsonic facility at Ohio State University.

In order to have confidence in the experimental set up, a series of tests were run with a simple 60-degree delta wing at a Reynolds number of 5×10^5 . This simple platform provided a means to collect data for verification by comparison to well documented thin delta wing's aerodynamic characteristics, [21, 22]. Expected aerodynamic characteristics for a low-speed 60-degree delta are: (1) the lift curve slope would be on the order of 0.05 per degree, (2) the stall angle-of-attack would be on the order of 35 degrees with a maximum lift coefficient of 1.3, and (3) the maximum lift-to-drag ratio, for a rounded leading-edge, would be on the order of 16 at a lift coefficient of 0.2. Similar testing and aerodynamic results have been verified at ARC's subsonic facility for a simple flat plate delta-type wing as validation for instrumentation and experimental setup [23].

CHAPTER 2. PLATFORM FOR TESTING AIRCRAFT MODEL

2.1 Wind Tunnel

Wind tunnel experiments were completed at ARC's 3'x5' low-speed Eiffel-type open circuit subsonic wind tunnel facility. The wind tunnel has a maximum airspeed of 180 ft/s, and a test section that is 39" high x 59" wide x 96" long. To run the tunnel, a LabView program was used that monitors the airspeed in real time. To determine the airspeed of the tunnel, a pressure differential between pressure ports was measured. The pressure differential was measured using static pressure ports at the end of converging section just ahead of the test section and via rings of surface pressure ports at the exit of the settling chamber [24].

2.2 Force Balance

To measure the aerodynamic loads of the wind tunnel model an HH-388 six-component internal force balance was used. Table 2 below provides important information of the balance and its calibration [25].

Table 2. Calibration Information [25]

Component	Max Design Load	Max Calibration Loads		Instrument Accuracy (% Load)
		Positive	Negative	
Normal Force 1	600 lbs	22.5 lbs	-22.5 lbs	-0.020
Normal Force 2	600 lbs	22.5 lbs	-22.5 lbs	-0.020
Side Force 1	600 lbs	22.5 lbs	-22.5 lbs	-0.020
Side Force 2	600 lbs	22.5 lbs	-22.5 lbs	-0.021
Rolling Moment	400 in-lbs	116.2 in-lbs	-116.2 in-lbs	-0.021
Axial Force	150 lbs	10.25 lbs	-16.9 lbs	-0.020

While the balance was calibrated on the lower end of the maximum design loads, the system still holds a high level of accuracy due to the design of the balance and the high gain, low noise signal conditioning and acquisition system [26]. The lift force and pitching moment are both measured with a combination of the two normal force components. Similarly, the side force and yawing moment are both measured with combinations of the side force components. Drag force and the rolling moment are measured with their own independent components. The balance can be seen below in Figure 6.



Figure 6. Six-Component Internal Force Balance

2.3 Sting Support System

To mount the model and balance in the tunnel, a steel parallelogram linkage system was used. The system was designed so that any vibration induced by unsteady aerodynamic flow would be damped out and not interfere with the measurement system [26]. A solid aluminum sting rod was used to mount the balance and model to the linkage system. In order to set the system at various angles-of-attack, a translating jackscrew was used to translate the system. As the jackscrew actuates up and down, the linkage system is rotated through a range of angles-of-attack, pivoted about the balance moment center [25]. To assist the jackscrew motor, a counterbalance weight system was used to reduce the load required to be moved by the motor. Modifications to the tunnel's ceiling were required in order to increase the maximum angle-of-attack of the system from 26 to 36 degrees. The expansion of the angle-of-attack range was required in order to capture the expected interesting aerodynamic characteristics of a delta wing at high angles-of-attack. Below, Figure 7, displays the system layout as previously described.



Figure 7. External View of Wind Tunnel with Installed Model

2.4 Data Acquisition and Reduction

Force balance data acquisition was acquired using a National Instruments Compact DAQ 9174. The system has a bit resolution of 32. Data was sampled at 2000 samples/second and averaged over a 5 second period for each data point. The data acquisition interface can be seen below in Figure 8. During testing, there were two manual

inputs the operator was responsible for, angle-of-attack and airspeed. The angle-of-attack was read from a calibration curve for the sting support system, and the airspeed was captured from the wind tunnel's LabView program.

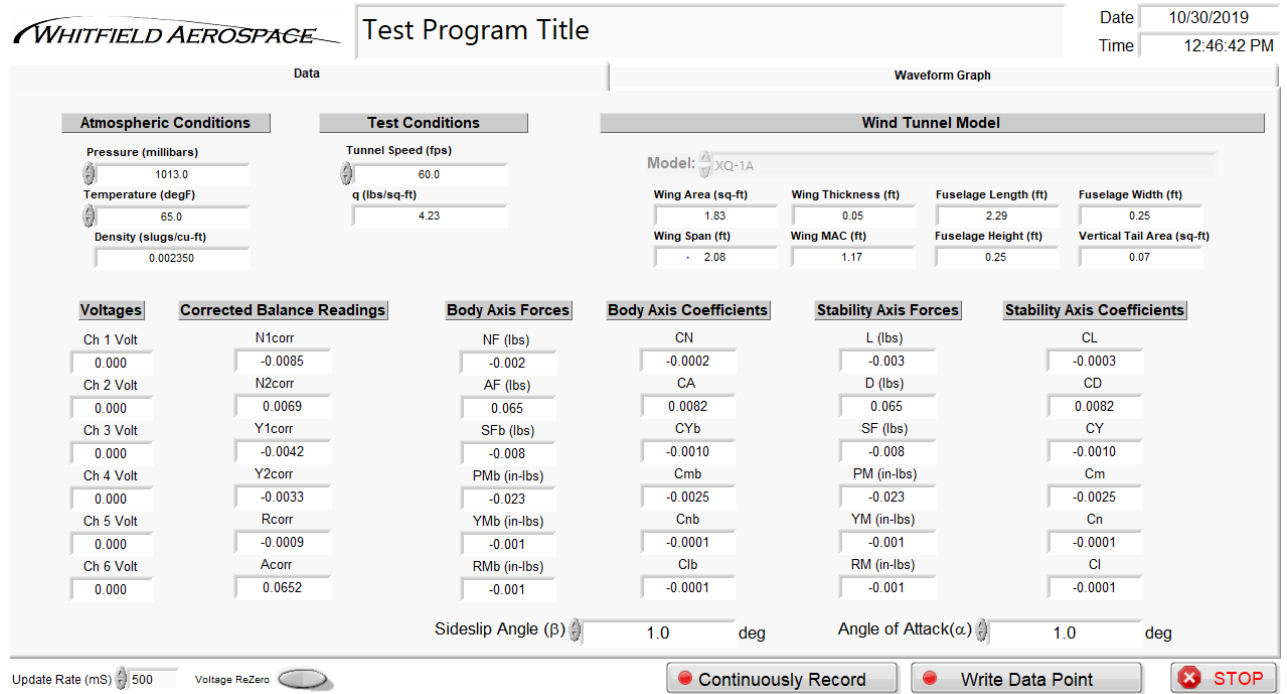


Figure 8. Screen Shot of Data Acquisition Interface

2.5 OpenVSP

OpenVSP is the computational program used to analyze the aircraft model. It is a panel method (PM) code created by the National Aeronautics and Space Agency (NASA),

and is shared publicly under the NASA Open Source Agreement (NOSA). Panel method codes are numerical tools that use the Prandtl-Glauert equation, Equation 4 [27], to modify the potential flow equation.

$$\nabla^2 \phi = (1 - M_\infty^2) \phi_{xx} + \phi_{yy} + \phi_{zz} = 0 \quad (4)$$

The ϕ symbol represents the perturbation velocity potential and M_∞ corresponds to free stream Mach number. Panel method codes model the flow around an aircraft by attaching singularity distribution structures (sources, vortices, and doublets) on each panel of a generated panel distribution that represents the aircraft's geometry. It was understood at the outset that OpenVSP would not capture the vortex component due-to-lift of a delta wing because panel methods only use potential flow theory. To account for the vortex effects in the OpenVSP results, the output data was modified with the leading-edge suction component, K_v , as detailed in section 1.1.1 above. In order to reflect the desired model, boundary conditions are applied to each panel, also known as control points [27]. Lomax, Heaslet and Fuller, [28], have defined the source, doublet and vortex potentials as follows:

$$\text{Source: } \phi_s = -\frac{Q}{r_c} \quad (5.a)$$

$$\text{Doublet: } \phi_d = \frac{Qz\beta^2}{r_c^3} \quad (5.b)$$

$$\text{Vortex: } \phi_v = -\frac{Qzv_c}{r_c} \quad (5.c)$$

In Equations 5a-c, Q represents the strength of the singularity structure. Q can be determined through boundary conditions defined by the flow field. Additionally, the remaining unknown variables from Equations 5a-c are defined as:

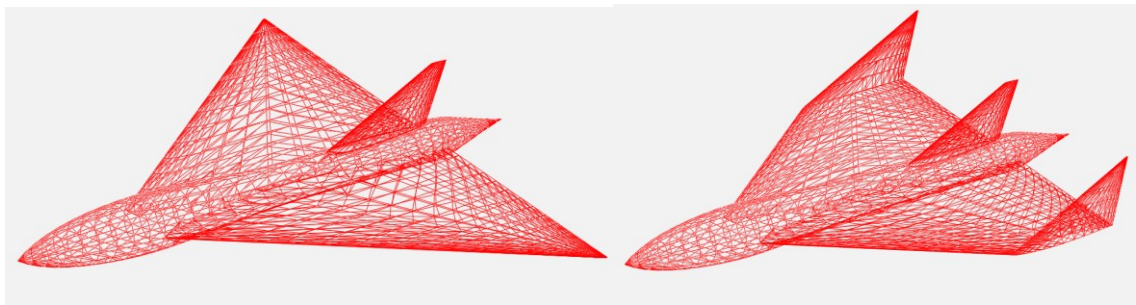
$$r_c = \sqrt{(x - x_1)^2 - \beta^2[(y - y_1)^2 + z^2]}$$

$$\beta^2 = M_\infty^2 - 1$$

$$v_c = \frac{x - x_1}{(y - y_1)^2 + z^2}$$

The x, y, and z variables are the coordinates for each singularity position over the aircraft panels. Variable r_c , is the hyperbolic radius which is the distance between two different singularities. By superimposing the singularity structures on each panel, the distribution ultimately satisfies the Prandtl-Glauert equation. Key limitations to using panel methods are that the configuration needs to be operating at low angles-of-attack so that the resulting flow can be assumed to be steady, inviscid, irrotational and incompressible [29]. Leading-edge suction theory and data from Zohar & Er-El [30] was used to verify the panel method code.

In Figure 9, the OpenVSP generated panel distribution for the analysis can be seen. Boundary conditions applied in the panel method code were set to match the boundary conditions used for wind tunnel testing. Two aircraft configurations were analyzed in OpenVSP, a baseline delta, and a positive 90-degree wing tip deflection delta wing layout. The results and discussion of the panel method analysis will be presented in Chapter 5 section 5.2, following discussions on the wind tunnel model, Chapter 3, and experimental procedure, Chapter 4.



a) Baseline Configuration

b) Positive 90-Degree Wing Tip
Deflection

Figure 9. OpenVSP Generated Panels for Aircraft Model

CHAPTER 3. AIRCRAFT WIND TUNNEL MODEL BUILDING PROCEDURE

3.1 Fuselage

As identified in Chapter 1, the fuselage has been area ruled for drag efficiency during high speed flight. The effect of area ruling can be seen in Figure 10 where the middle of the fuselage body narrows. In order to mount the model to the internal balance, an aluminum collar was required. The collar was positioned inside of the fuselage body so that the balance can be pinned at the model's center of mass. To hold the collar in-place, foam inserts were used to position the balance collar along the fuselage's centerline. The body of the fuselage was manufactured in two symmetrical halves. Each half was made with a fiberglass epoxy layup. Each layer of epoxy and fiberglass sheets were laid into a female foam mold that held the outer fuselage shape. Once every layer of epoxy and fiberglass were set into the mold, the mold and layup were then placed into a vacuum sealed bag for curing. After each side of the fuselage was cured, they were glued together with high strength adhesive. The final steps for finishing the fuselage was sanding and painting until the surface was smooth. Figure 10 below shows the fuselage without wings and the exposed simple spar layout. With the simple spar layout displayed, any wing platform can be installed on this fuselage with the only limiting factor being the wing root thickness.

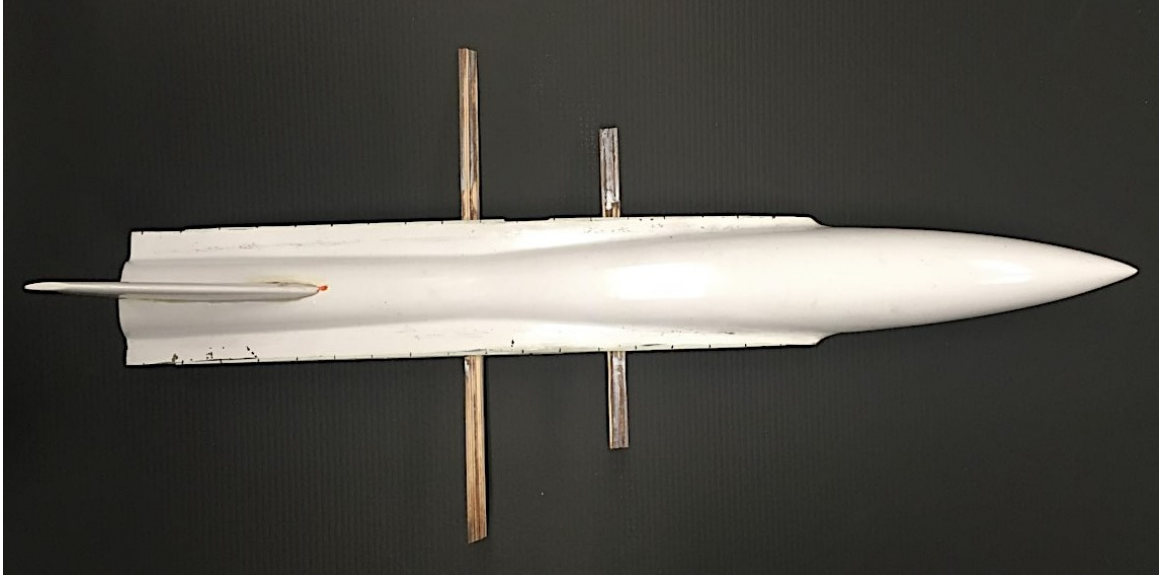


Figure 10. Wind Tunnel Model Fuselage

3.2 Delta Wing

The aircraft model that was under analysis had a novel delta wing with a NACA 64-006 series airfoil that produces a rounded leading-edge, a 60-degree wing sweep, and rotating wing tips. The airfoil was chosen for its high-speed capabilities and the resulting interesting leading-edge vortex effects from the rounded leading-edge. As stated in Chapter 1, the leading-edge sweep of 60 degrees was chosen due to the depth of previous research on the platform. Earnshaw and Lawford studied how the leading-edge sweep of delta wings impact the lift coefficient, as seen in Figure 11 [31]. Figure 11 shows how the 60-degree delta has the third largest maximum lift coefficient while also having one of the steepest lift curve slopes.

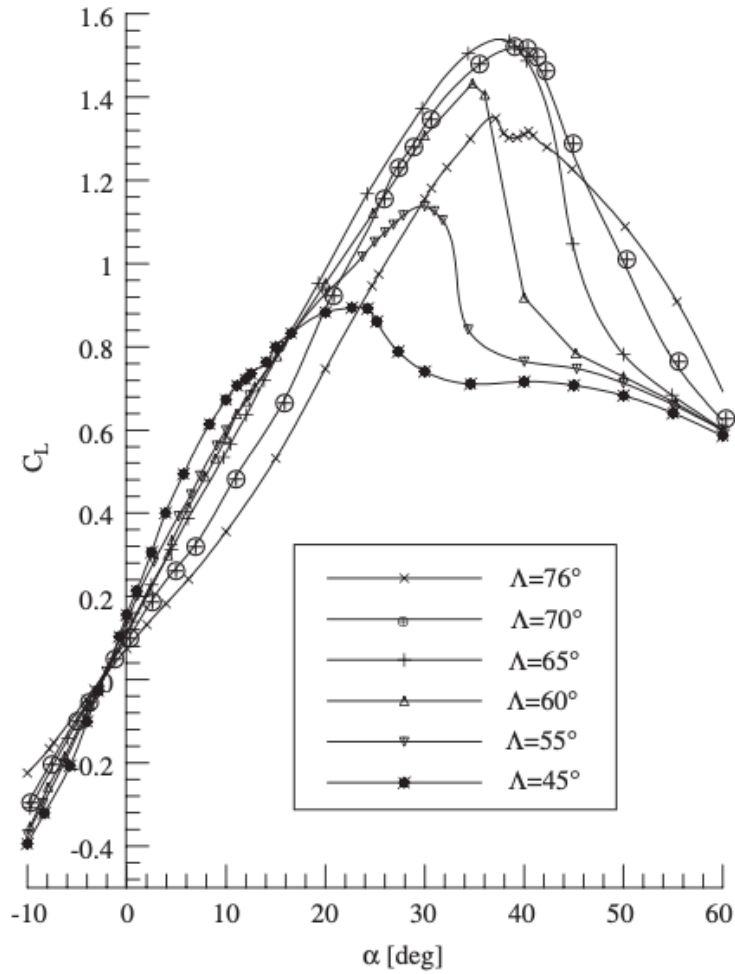


Figure 11. Variation of lift coefficient with angle-of-attack [31].

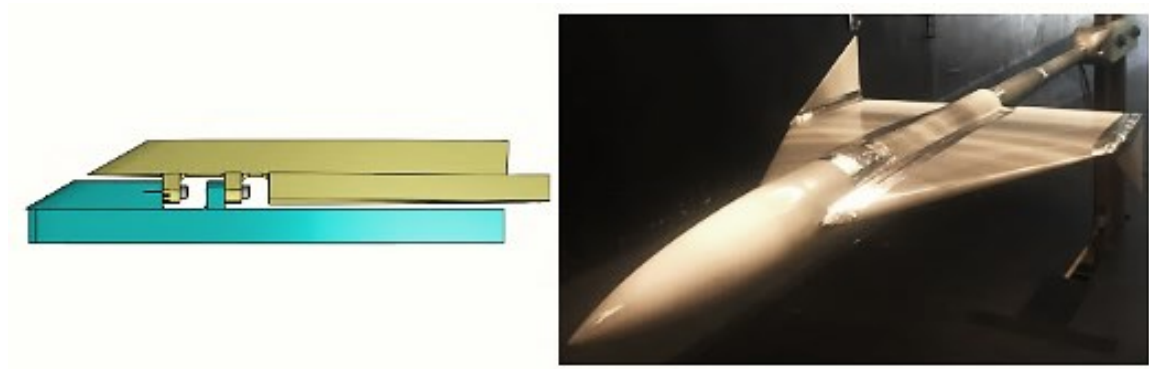
Manufacturing the delta wing model followed the same procedure as the fuselage body described above. Two female foam molds of the delta wing were wrapped with the fiberglass epoxy layup and vacuum bagged for curing. Each side of the wing was created

in two parts, a top and bottom, and was fixed together with high strength glue. The root chord of the delta wing was made from laser cut plywood in the shape of the NACA 64-006 series airfoil, and then glued into the root of the hollow wing body. In order for the fuselage spars to mate with the wing, the root chord had two corresponding channels that fit together with the fuselage spars. The wing spar channels and fuselage spar configuration resulted in a secure fit between the two components. Once the delta wing was fit into place on the fuselage, metal tape was used along the delta wing and fuselage junction in order to cover any small gaps between the two parts. The wings were also finished with extensive sanding and painting to achieve a smooth surface finish. The final step for completing the wings was to cut a one-inch strip centered at $2/3^{\text{rds}}$ half-span on each wing for installation of the rotating wing tip mechanisms.

3.3 Wing Tip Hinge Mechanism

The delta wing used in this study had wing tips that can articulate about a hinge in the wing at $2/3^{\text{rds}}$ half-span, parallel to the aircraft's fuselage. In order to rotate the wing tips a custom hinge had to be designed. The articulating wing tips were desired to be able to be deflected 180 degrees, 90 degrees up and down. The sign convention used for the wing tip deflection orientation is, positive when the wing tips are deflected up and negative when the wing tips deflected downward. The hinge was designed to fit snugly into the hollow wing so that the flow around the wing was not perturbed by an external mechanism. The hinge mechanism was manufactured using 3D printing and consisted of two parts per each side of the wing. Each component of the mechanism was inserted into the wing body

and fixed in place with high strength glue. Two cap screws enable the wing tips to be locked in place at a prescribed wing tip angle. Figure 12, below, shows the computer-aided design (CAD) model of the hinge (Figure 12 a), and the installed hinge at its asymmetrical deflected extremes (Figure 12 b) of plus and minus 90 degrees.



a) CAD Model of Hinge

b) Wing Tips Deflected to Hinge Extremes in
the 90° Asymmetric Configuration

Figure 12. Wing Tip Deflection Mechanism

3.4 Flow Visualization

Tufts were used to observe the behavior of the flow over the delta wing. The tufts were arranged in a one by one-inch grid and each tuft was two inches long. The material used for the tufts was neon orange sewing thread. White art tape was used to adhere the tufts to the wing's surface. Other tapes were considered like duct tape, double sided tape

with the wrapper left on one side, and scotch tape, but the art tape was the best option. Art tape struck a balance between adhesive strength, opacity, and thickness. It was strong enough to stay on during the wind tunnel tests, opaque enough not to show the thread taped underneath, and it was thin enough to not drastically change the surface of the wing.

CHAPTER 4. EXPERIMENTAL PROCEDURE

4.1 Measurement Frames of Reference

The forces measured by the 6-component internal force balance are in the body axis frame of reference and are converted into the stability axis frame of reference. The sign convention for the body axis and stability axis frames can be seen in Figure 13 and Figure 14. The body axis reference frame moves with the rotation of the sting support system and model. The stability axis frame changes as the model is rotated in order to have the lift and drag forces of the model perpendicular and parallel to the flow direction, respectively. When the model is in a sideslip angle, β (degrees), the drag force is measured in line with the model's centerline. Body and stability reference frame conversion equations can be seen in Table 3 below. These conversions are built into the data acquisition system, and both frames are observed in real time during the test on the data acquisition interface, per Figure 8.

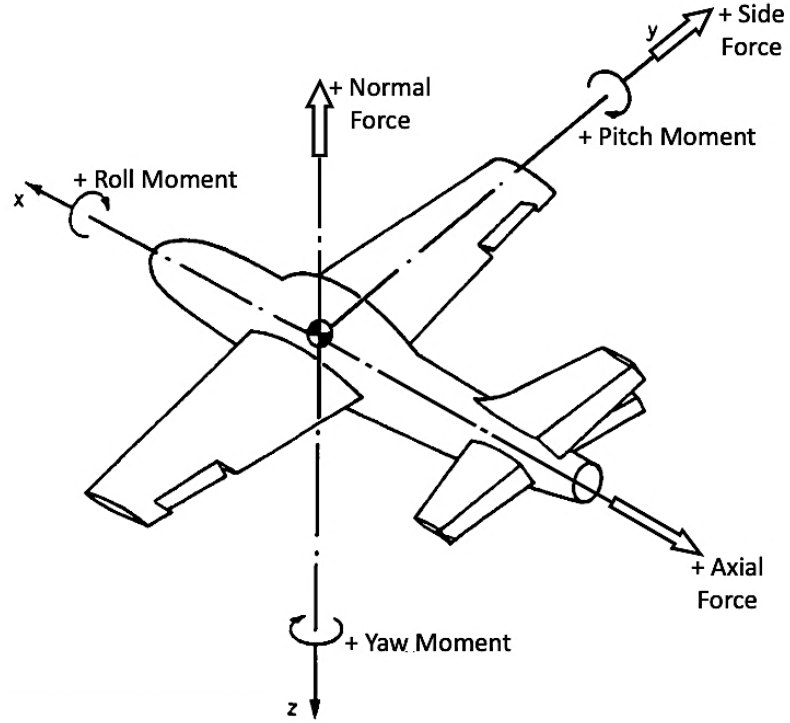


Figure 13. Body Axis Frame of Reference [32]

Table 3. Equations for Converting from Body (_B) to Stability Axis (_{SA}) Frame of Reference [33]

$LF_{SA} = NF_B \cos \alpha - AF_B \sin \alpha$	$RM_{SA} = RM_B \cos \alpha + YM_B \sin \alpha$
$DF_{SA} = AF_B \cos \alpha + NF_B \sin \alpha$	$YM_{SA} = YM_B \cos \alpha - RM_B \sin \alpha$
$SF_{SA} = SF_B$	$PM_{SA} = PM_B$

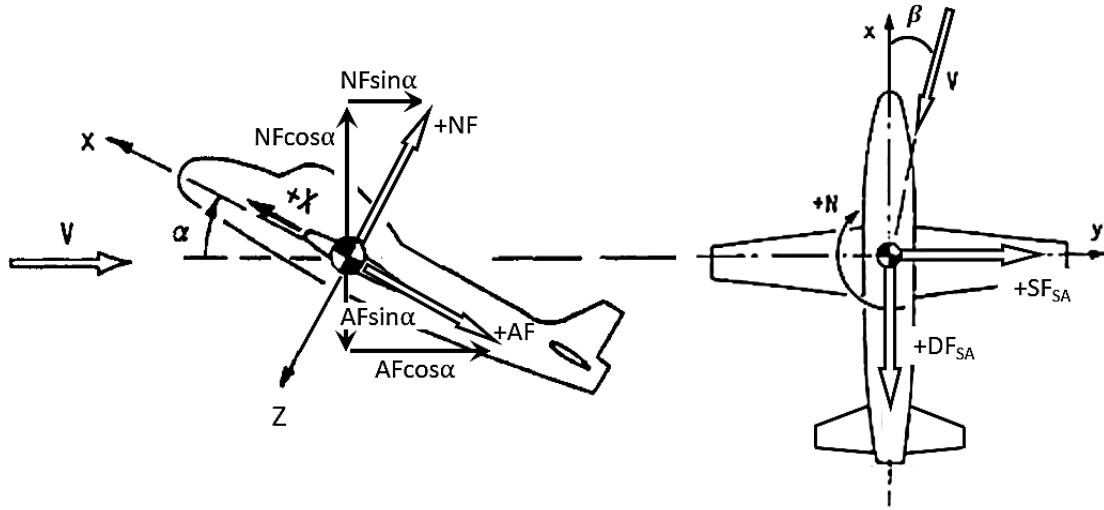


Figure 14. Stability Axis Frames of Reference [32]

4.2 Tunnel Corrections and Experimental Considerations

The corrections detailed below are applied to the wind tunnel data and accounted for within the data acquisition system so that all values observed in real-time during the testing had the corrections automatically applied.

4.2.1 Flow Constraint / Tunnel Blockage

When testing an aircraft model in a wind tunnel, the flow around the model accelerates more than it would in free-air conditions. This effect is due to the solid blockage from the installed model. Following Bernoulli's equation and continuity, flow around the model accelerates due a decrease in test section area in the region of the model and test

section walls. The increase in velocity is assumed to be constant over the aircraft. To help mitigate the blockage, it is recommended to have a ratio of the frontal area of the test article to wind tunnel cross sectional area between 1 and 10% [33]. The model used here has a blockage ratio of about 1.5% and 7.8% at 0 and 35 degrees angle-of-attack, respectively. To correct for the increase in velocity, equation (6) was used [33]. In the equation below, V is the tunnel's unchanged airspeed and u_1 is the average change in tunnel velocity due to the model's size. The remaining variables are the volume of the model (Vol) and the volume of the tunnel test section. The variable h represents the length of the test section wall perpendicular to the wingspan and b represents the length of the wall parallel to the wingspan.

$$\frac{u_1}{V} = 0.62 * \frac{Vol}{h^2 b} \quad (6)$$

$$Vol = 0.70(Model\ Thickness)(Wing\ Span)(Root\ Chord)$$

4.2.2 Downwash Correction / Closed Octagonal Jet

To correct for the reduced drag measured in a closed jet, equation (7) was used [33]. Drag is reduced in a wind tunnel due to the induced flow in the lift direction being altered by the close proximity of the test section walls. Another issue with a closed test section with boundaries close to the model is with the increase of the local angle-of-attack along the span of the wing. The local angle-of-attack is corrected using equation (8) [33]. In a closed test section, tip stall can occur at lower geometric angles-of-attack due to the

increased local angle-of-attack. If the wing tips are too close to the test section walls, the wing tip vortices can be affected by the walls. This effect can be mitigated by ensuring the model's span is no more than 80% of the test section width [33]. The model under test has a wingspan of two feet, which corresponds to 42% of the test section width, thereby diminishing the impact of the walls on the wing tip vortices. In the equations below, δ is the boundary correction factor as defined by Barlow et al. [33], S is the wing area and C is the tunnel jet cross-sectional area.

$$\Delta C_D = \delta \frac{S}{C} C_L^2 \quad (7)$$

$$\Delta \alpha = \delta \frac{S}{C} C_L \quad (8)$$

4.2.3 Model Base Drag

Equation (9) is used to account for the added drag due to the open cavity at the back of the model that is required for installation of the internal force balance and sting. This force contributes to the axial body force, and needs to be removed from both the lift and drag measurements. The base drag is also known as pressure drag, which is the result of the differences in pressures of the freestream and in the cavity. In equation (9) [26], $D_{fuselage}$ and D_{sting} are the diameters of the aft fuselage and sting, respectively, and q_∞ is the dynamic pressure.

$$C_{D_{Base}} = \frac{q_\infty}{S} \pi \left(\frac{D_{fuselage} - D_{sting}}{2} \right)^2 \quad (9)$$

4.2.4 Weight Tare

In order to measure the aerodynamic forces of the model, the model's weight needed to be determined through the angle-of-attack range. The model's weight was found by collecting data from the internal force balance with no external loads (wind tunnel off and doors closed) throughout the angle-of-attack range. It is necessary to take weight measurements for the full angle-of-attack range, because as the model is rotated, different components of the internal force balance are impacted by the weight of the model. For every subsequent aerodynamic load test, the model's weight values were accounted for with each corresponding angle-of-attack.

4.3 Run Procedure

Before every series of tests, the wind tunnel was zeroed to account for the current day's ambient conditions with the tunnel doors closed. When performing the tests, two people were required, one to operate the data acquisition software and the other to monitor the airspeed, angle-of-attack, and model in the tunnel. Tests were performed at two Reynolds numbers, 3.65×10^5 and 5×10^5 , based on the mean aerodynamic chord. The model was stepped from 0-36 degrees angle-of-attack, with data being recorded every two degrees. The process explained above was repeated for the flow visualization tests apart from not recording balance data. Two people were still required to run the tests, but the data acquisition system operator worked as the camera operator instead.

4.4 Range of Testing

In order to assess the complete effect that the deflected wing tips have on the aerodynamic performance of the aircraft, numerous wing tip deflection configurations were tested. For each wing tip deflection direction (positive, negative and asymmetric), four tests were performed with varying angles-of-attack. Each test corresponded to a ± 30 , ± 45 , ± 60 , and ± 90 -degree wing tip deflection. Each model wing tip configuration was swept through an angle-of-attack range of 0-36 degrees. The high angle-of-attack boundary was set so that the stall condition of the delta wing would be captured. The baseline configuration was used in two separate tests: (1) an angle-of-attack sweep and (2) a sideslip sweep while held at a constant angle-of-attack of 0 degrees. The data examined here has been gathered from a total of 18 wind tunnel tests.

CHAPTER 5. RESULTS AND ANALYSES

5.1 Overview

The following sections detail the panel method analysis results and the wind tunnel data for the novel delta wing at the center of this study. The panel method results and how they compare to theory and known data will be covered first. Next, the panel method results will be compared to wind tunnel data for the aircraft under discussion. After the panel method discussion, the data gathered with the 3'x5' wind tunnel at ARC will be presented in the following order: lift and drag tests, longitudinal tests, lateral and directional tests, Reynolds number effects, and the discussion of flow visualization images. The four different delta wing configurations with a wing tip deflection of 60 degrees can be seen in Figure 15 below.

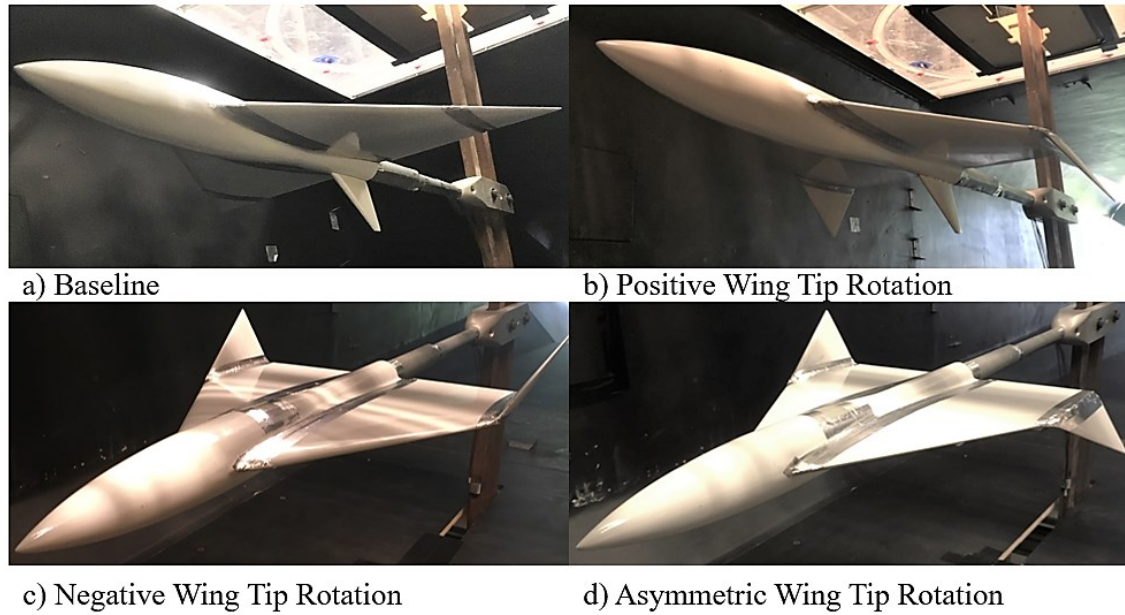


Figure 15. Four Delta Wing Configurations with 60° Wing Tip Deflection

5.2 Panel Method Results

5.2.1 Panel Method Results vs Theory and External Data

The panel method code was found to accurately predict the lift curve slope of the delta wing, after implementing the vortex lift component, as seen in Figure 16. The percent difference of lift curve slope between the panel method data with the leading-edge suction correction and the leading-edge suction analogy is 2.2%. The panel method data does not accurately follow theory, which is as expected because the panel method analysis only solves the potential flow model and does not account for vortex lift. The symbol α , in Figure 16, represents the angle-of-attack in degrees.

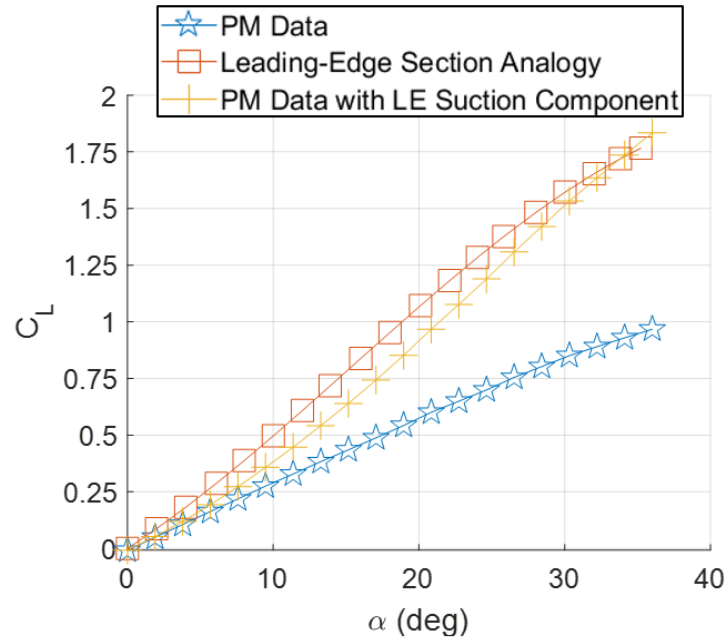


Figure 16. Panel Method (PM) Data with and Without the Leading-Edge Suction Analogy [6, 8].

While the adjusted panel method data follows theory, it was not able to accurately predict the pitching moment. The two panel method data curves presented in Figure 17 are unadjusted and adjusted with the leading-edge suction theory. Figure 17 also presents pitching moment data from Zohar and Er-El that was found through a surface pressure study of a 60-degree delta with an aspect ratio of 2.3 [30]. The delta in the Zohar and Er-El study closely matches the delta under analysis, with both sharing the 60-degree sweep. As seen below in Figure 17, the unadjusted panel method data diverges from previous data. The data from Zohar and Er-El has a pitching moment slope of -0.0298, while the

unadjusted panel method pitching moment slope is -0.0103. The error between the external data and the unadjusted panel method result is in 65% error.

By using equations (10) and (11), an adjustment for nonlinear vortex lift component of the pitching moment can be applied to the panel method data. Equation (10) has the panel method data pitching moment ($C_{m_{PM}}$), which corresponds to the linear contribution of the pitching moment, and the nonlinear contribution of pitching moment ($C_{m_{Nonlinear}}(\alpha)$) as a function of angle-of-attack. A similar method in separating the pitching moment into the linear and nonlinear components was performed by Kirby [34]. In equation (11), $C_{L_{Vortex Lift}}(\alpha)$ is the coefficient of lift due to the nonlinear vortex lift as a function of angle-of-attack [6,8], and $\frac{x_{cg}-x_{ac}}{\bar{c}}$ is the moment arm of the lift force. The variables of the moment arm will be detailed below in section 5.4. The key assumption in using this method is estimating the vortex lift to be acting at the aerodynamic center of the model. The adjusted pitching moment slope is -0.0161, which results in 46% error when compared to the external data. Even though the adjusted panel method data has a lower percent error than the unadjusted (46% < 65%), the adjusted panel method data still diverges from the expected pitching moment data.

$$C_{m_{adjusted}} = C_{m_{PM}} + C_{m_{Nonlinear}}(\alpha) \quad (10)$$

$$C_{m_{Nonlinear}}(\alpha) = C_{L_{Vortex Lift}}(\alpha) \left(\frac{x_{cg}-x_{ac}}{\bar{c}} \right) \quad (11)$$

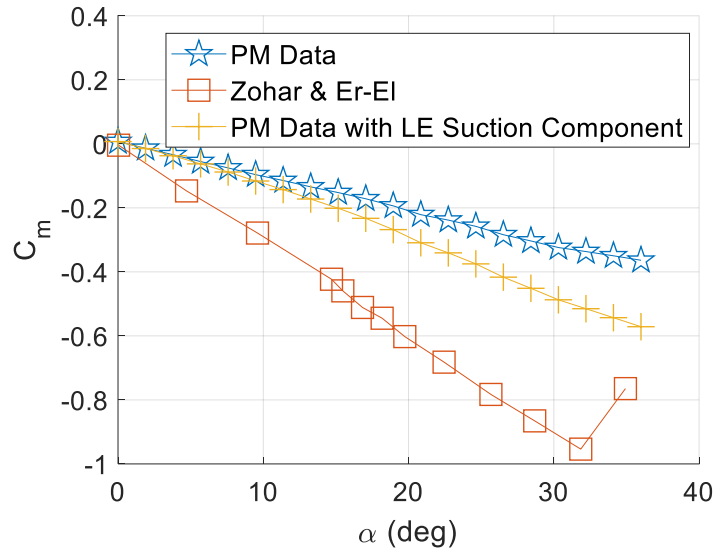


Figure 17. Pitching Moment Comparison Between unadjusted and adjusted Panel Method (PM) Baseline Configuration Data and Data from Zohar and Er-El [6, 8, 30]

5.2.2 Panel Method Results vs Wind Tunnel Data

The panel method data compared to the wind tunnel data is presented below in Figure 18. The panel method data in the lift curve plot, Figure 18 a), has been adjusted with the leading-edge vortex lift component. The trend observed above in which the panel method lift curve slope agrees with the expected lift curve slope is repeated here. The lift curve slope for the baseline plots agree closely, but the 90-degree delta wing layout data diverges at high angles-of-attack (25 degrees and up). The divergence between the panel method data and wind tunnel model for the 90-degree wing tip deflection configuration could be attributed to a reduction of vortex lift of the wind tunnel model. The data may

imply that a smaller leading-edge, due to a portion of the leading-edge being deflected positive 90 degrees, results in a weaker leading-edge vortex, which in turn produces less vortex lift. Figure 18 b) shows how the adjusted panel method data poorly predicts the pitching moment of the wind tunnel model, just as was observed in the section above.

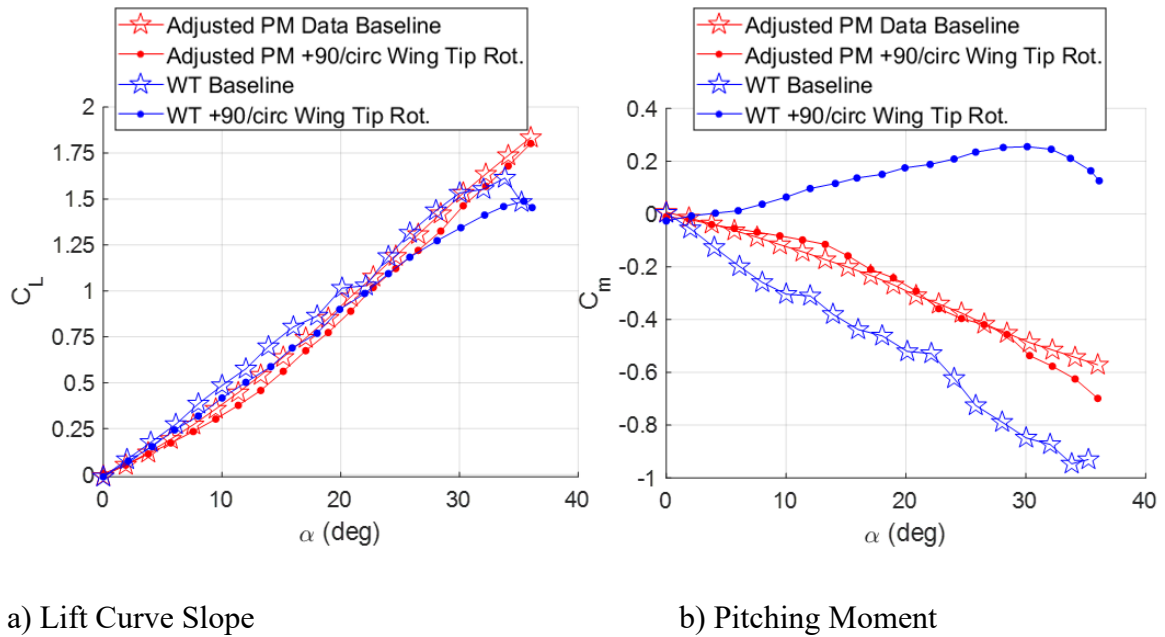


Figure 18. Panel Method (PM) Results Compared to Wind Tunnel (WT) Results [6,8]

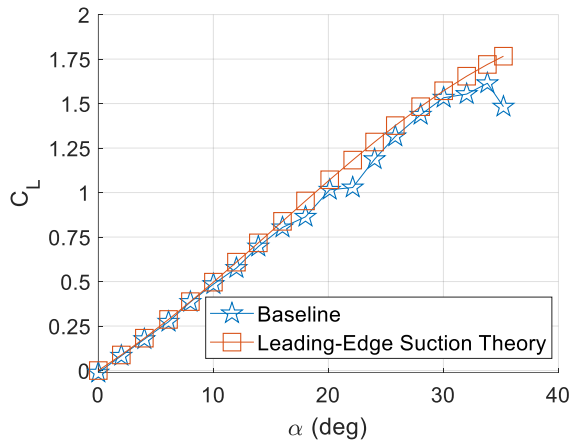
5.3 Lift/Drag Tests

5.3.1 Baseline Delta Wing

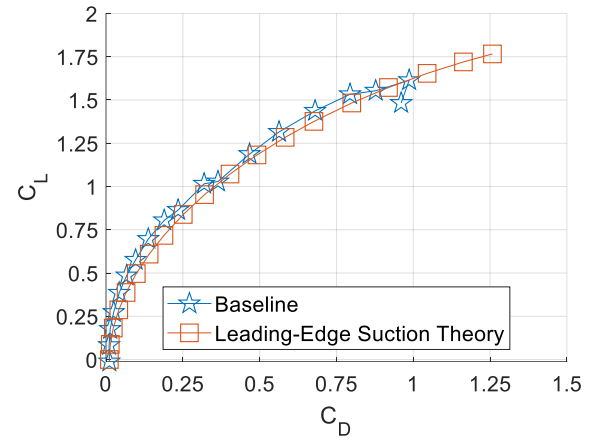
The model data follows the predicted theory closely up until large angles-of-attack with a 4% difference in lift curve slopes, Figure 19 a). The lift curve slope of the leading-edge suction theory and baseline model are, 0.0501 and 0.0481, respectively. The divergence between theory and model data at stall is expected because of the model's rounded leading-edge compared to the sharp leading-edge used in analytical estimations, and the theory does not hold when vortex instability and separation occur. Leading-edge suction theory and model data completely diverges when stall occurs at an angle-of-attack of 34 degrees. Additionally, the baseline lift curve slope follows the expected trends laid out in reference [21]. The lift curve slope of a 60-degree leading-edge delta wing is expected to be 0.05 per degree while the baseline has a lift curve slope of 0.0501 per degree. Maximum lift coefficient is expected to be on the order of 1.3 at an angle-of-attack of 35 degrees while the baseline has a maximum lift coefficient of 1.614 at a 33.8-degree angle-of-attack. The difference in maximum lift coefficients can be attributed to the geometric differences between the model's airfoil and the flat plate used by the analytical estimations.

The drag polar, Figure 19 b), shows the strong similarity between theory and the baseline model. The main difference between theory and the model in the drag polar is that the model stalls before theory which results in the divergence between the two curves around a drag coefficient of 1.0. The aerodynamic efficiency, Figure 19 c), accentuates the minor differences between theory and the model. The maximum lift-to-drag ratio of 12.9

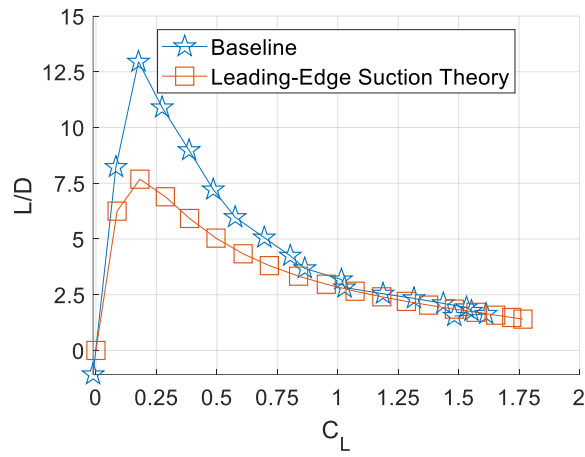
for the model is much higher than theory, 7.68, due to the variations in drag between theory and experimental results at low lift coefficients and angles-of-attack. The drag coefficients at the maximum lift-to-drag ratio for theory and the model are, 0.0237 and 0.0136, respectively. While the lift coefficients at the maximum lift-to-drag ratio for theory and the model are, 0.1818 and 0.1760, respectively.



a) Lift Curve Slope



b) Drag Polar



c) Aerodynamic efficiency

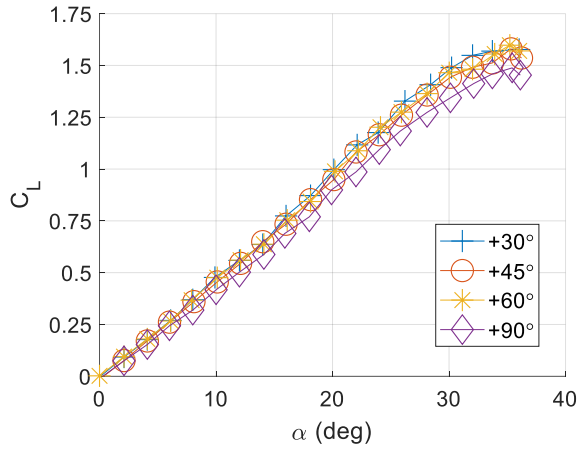
Figure 19. Comparison between Leading-Edge Suction Theory and Wind Tunnel

Data [6, 8]

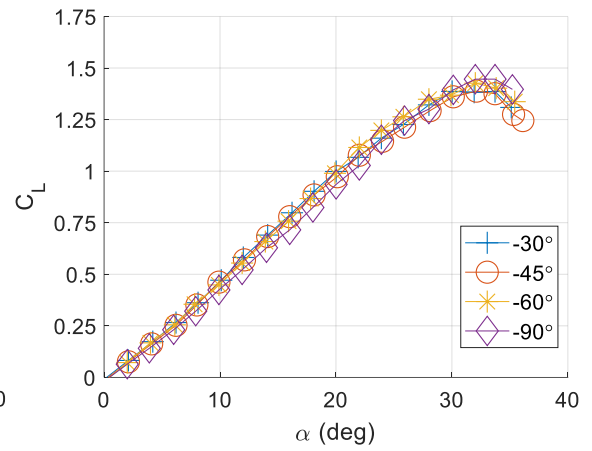
5.3.2 Deflected Wing Tip Delta Wing

The lift curve slope of the positive wing tip layout, Figure 20 a), is largely unchanged with wing tip deflection until the 90-degree wing tip deflection. The 90-degree wing tip layout has a similar lift curve slope to the other layouts but has a visibly lower maximum lift coefficient of 1.49. The average maximum lift coefficient for the 30 through 60-degree wing tip deflections is 1.59. The negative wing tip configuration, Figure 20 b), does not show any significant change in lift curve slope with wing tip deflection. The average lift curve slope and average maximum lift coefficient for the negative wing tip configuration is 0.045 per degree and 1.41, respectively. The asymmetric wing tip configuration, Figure 20 c), does show slight variations in lift curve slope with increased wing tip deflection, unlike the previous configurations. The minor trend for the asymmetric lift curve slope is that as the wing tip deflection angle increases, the lift curve slope and maximum lift coefficient decreases.

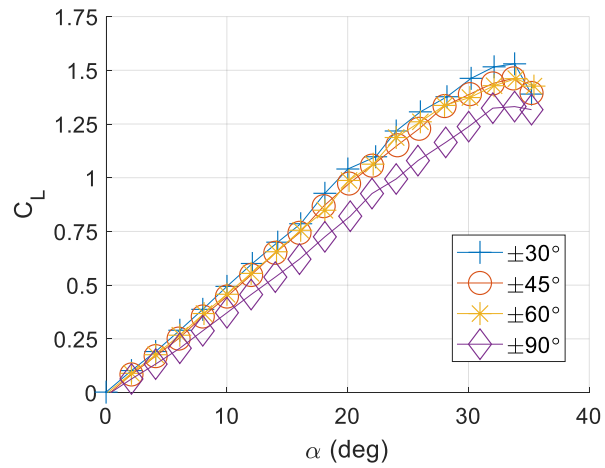
Figure 21 shows the drag polar for the three wing tip configurations. The drag polar for the positive and negative layouts, Figure 21 a) and b), show minor changes with increasing wing tip deflection. While the asymmetric drag polar, Figure 21 c), shows a decrease in maximum lift and drag values with an increase in wing tip deflection. The coefficient of drag at zero lift for the positive, negative and asymmetric configurations are 0.01.



a) Positive Wing Tip Deflection

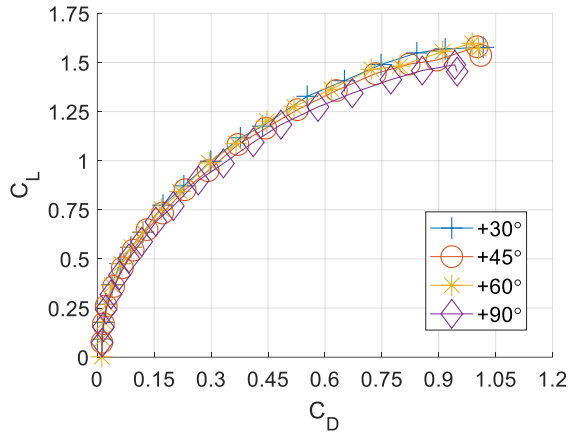


b) Negative Wing Tip Deflection

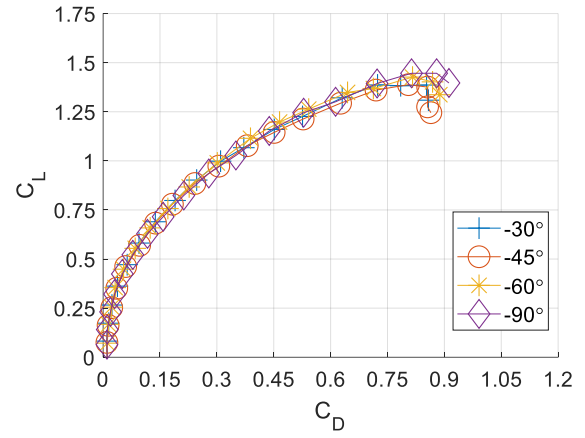


c) Asymmetric Wing Tip Deflection

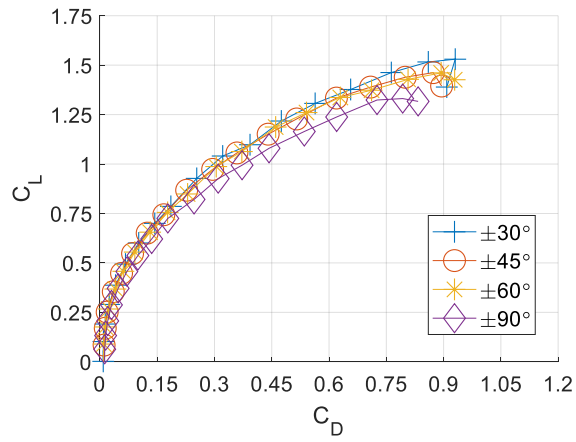
Figure 20. Various Wing Tip Deflection Lift Curve Slope Plots



a) Positive Wing Tip Deflection



b) Negative Wing Tip Deflection

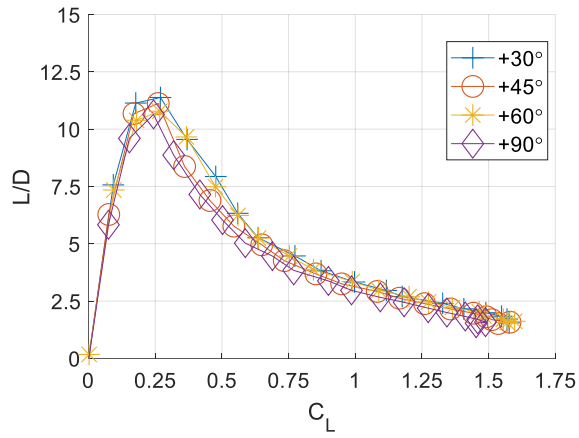


c) Asymmetric Wing Tip Deflection

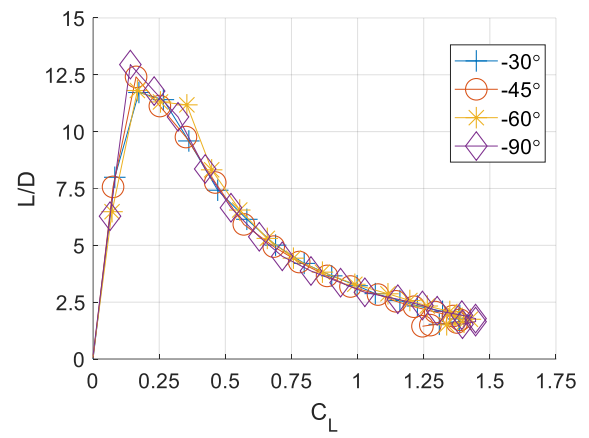
Figure 21. Various Wing Tip Deflection Drag Polar Plots

The aerodynamic efficiency of the positive wing tip deflections, in Figure 22 a), shows that as wing tip deflection increases, and the maximum aerodynamic efficiency marginally decreases. Of the positive wing tip deflections, the maximum lift-to-drag is 11.38 at a 0.270 lift coefficient for the 30-degree layout. The drag that corresponds to the maximum lift-to-drag ratio of the 30-degree layout is 0.0148. With increasing positive wing tip deflection, the wing tips function as winglets. The wing tips, while deflected in the positive direction act as a wall for the vortex as it forms and breaks with higher angles of attack. It is known that winglets act as a wall between the top and bottom of the wing by preventing the high-pressure air on the underside of the wing from mixing with the low-pressure air on the top of the wing [20].

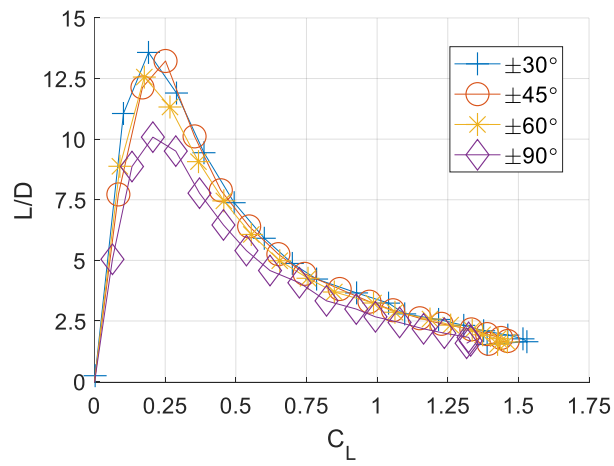
The negative wing tip deflection has a marginal increase in maximum lift-to-drag ratio with increased wing tip deflection, which is opposite to the trend observed for the positive wing tip deflection. The maximum lift-to-drag of the negative deflection layouts is 12.95 at a lift coefficient of 0.141, Figure 22 b), at a wing tip deflection of 90 degrees. The coefficient of drag associated with the maximum lift-to-drag is 0.0109 and can be seen in Figure 21 b) near the zero lift drag coefficient of 0.01. When the wing tips are in the negative configuration, they act as drooped winglets by preventing mixing of the low and high-pressure air from the top and bottom of the wing, respectively. The maximum lift-to-drag of the asymmetric wing tip layout is 13.58 at a lift coefficient of 0.191 and a deflection of 30 degrees, Figure 22 c). The corresponding drag coefficient at the aforementioned lift-to-drag ratio is 0.0190, which is near the zero lift drag coefficient, 0.01, in Figure 21 c).



a) Positive Wing Tip Deflection



b) Negative Wing Tip Deflection



c) Asymmetric Wing Tip Deflection

Figure 22. Various Wing Tip Deflection Aerodynamic Efficiency Plots

5.3.3 Comparison between Configurations

Figure 23 compares the 90-degree wing tip deflection for all configurations to the baseline. The positive and negative deflected wing tips closely matched the lift curve slope of the baseline until high angles of attack. While, the asymmetric layout was consistently lower. This trend was repeated for the drag polar, Figure 23 b). This trend of divergence between wing tip deflection and the baseline could be attributed to a weaker leading-edge vortex on the deflected wing tip deltas, because there is less of a continuous leading-edge to support the vortex. The aerodynamic efficiency curves of the positive and asymmetric wing tip deflection are consistently lower than the baseline for low angles-of-attack. On the other hand, the negative wing tip deflection follows the baseline aerodynamic efficiency closely for all angles-of-attack, Figure 23 c). The difference between the positive and negative maximum lift-to-drag ratio could imply that the negative wing tip layout is a more efficient wing tip configuration because it has a higher aerodynamic efficiency than the positive wing tip configuration. A summary of the significant aerodynamic values in Figure 23 can be found in Table 4 below.

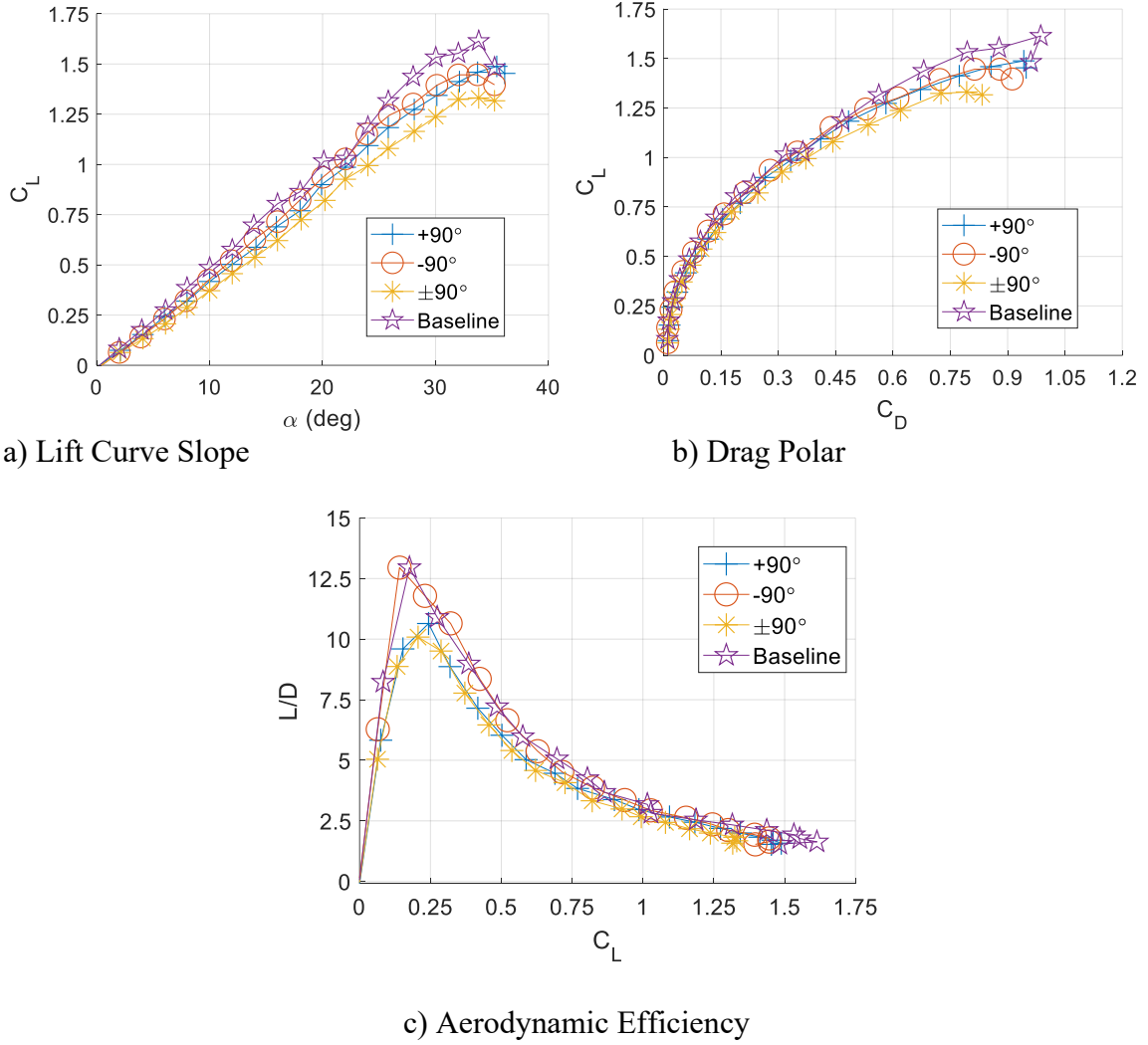


Figure 23. Comparison of Lift and Drag Data Across Each Configuration for a Wing Tip Deflection of 90°

Table 4. Lift Characteristics of Each 90° Wing Tip Deflection Configuration

	Positive	Negative	Asymmetric	Baseline
$C_{L \max}$	1.488	1.446	1.331	1.614
$C_{L\alpha}$	0.0423	0.0435	0.0398	0.0481
L/D	10.64	12.95	10.08	12.95

5.4 Longitudinal Tests

The key requirements for balanced static longitudinal stability are a negative pitching moment slope and a positive pitching moment coefficient at an angle-of-attack of 0 degrees. A longitudinally stable aircraft will produce a corrective pitching moment, as a result of a pitch perturbation, that results in level flight.

Equation (12) [35] solves for the neutral point where X_{np} is the neutral point, X_{ac} is the aerodynamic center, $\bar{c}$ is the mean aerodynamic chord, $C_{m\alpha_f}$ is the fuselage pitching moment, $C_{L\alpha_w}$ is the lift curve slope of the wing, V_H is the horizontal tail volume ratio, η is the ratio of tail to wing dynamic pressure, $C_{L\alpha_t}$ is the lift curve slope of the horizontal tail, and $\frac{d\varepsilon}{d\alpha}$ is the change of downwash angle with respect to angle-of-attack. When calculating the neutral point for the wind tunnel model, the third component in equation (10), $\eta V_H \frac{C_{L\alpha_t}}{C_{L\alpha_w}} \left(1 - \frac{d\varepsilon}{d\alpha}\right)$, goes to zero because there is no horizontal tail on the model ($V_H = 0$). The lift curve slope of the wing is the lift curve slope of the baseline model (0.0481 per degree) and the fuselage pitching moment (0.003 per degree) was found using equation (13), from NASA TR 711 [36].

In equation (13), K_{fus} is an empirical fuselage pitching moment factor, W_f is the maximum width of the fuselage, L_f is the length of the fuselage, and S_w is the wing area. The neutral point is 16.05 inches from the nose of the wind tunnel model. It will be assumed that the neutral point is equal to the aerodynamic center (16.95 inches from the nose) due to the similarity in the calculated neutral point and the aerodynamic center (only a 5% difference). The aerodynamic center of the wing was calculated at 26% percent of the mean

aerodynamic chord, which corresponds to the aerodynamic center for a NACA 64-006 airfoil [37]. The static margin is shown as equation (14) [35]. For a stable aircraft, the static margin is required to be positive. Figure 24 identifies the aerodynamic center and center of gravity on the model. The model has a positive static margin due to the aerodynamic center dimension being larger than the center of gravity dimension. Figure 25 shows how static longitudinal stability changes as the static margin of the wind tunnel model changes.

$$\frac{X_{np}}{\bar{c}} = \frac{X_{ac}}{\bar{c}} - \frac{C_{m\alpha_f}}{C_{L\alpha_w}} + \eta V_H \frac{C_{L\alpha_t}}{C_{L\alpha_w}} \left(1 - \frac{d\varepsilon}{d\alpha}\right) = \frac{X_{ac}}{\bar{c}} \quad (12)$$

$$C_{m\alpha_f} = \frac{K_{fus} W_f^2 L_f}{\bar{c} S_w} \quad (13)$$

$$\text{Static Margin} = \frac{X_{ac}}{\bar{c}} - \frac{X_{cg}}{\bar{c}} \quad (14)$$

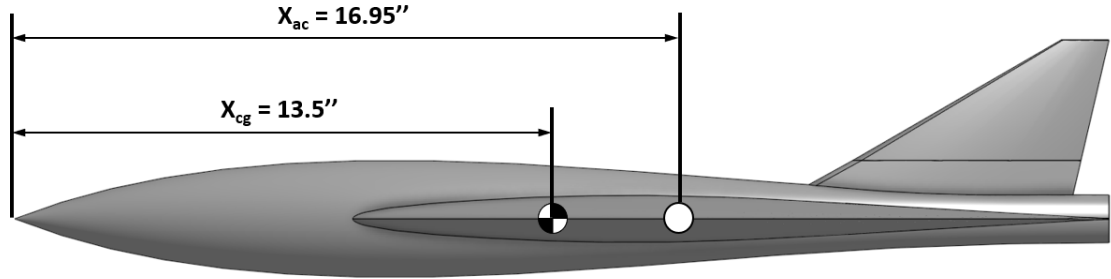


Figure 24. Baseline Model with Center of Gravity and Aerodynamic Center Shown

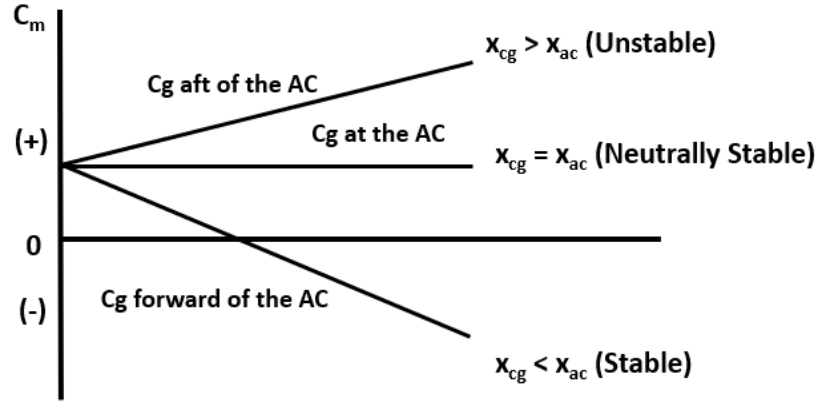


Figure 25. Static Longitudinal Stability [35]

5.4.1 Baseline Delta Wing

Figure 26, below, identifies the pitching moment similarity between the wind tunnel model and the Zohar and Er-El study. The pitching moment slope for Zohar and Er-El is -0.0298 while the baseline pitching moment slope is -0.0280 (6% difference) [30]. The baseline configuration is statically stable with an aerodynamic center aft of the center of mass, Figure 24.

To further investigate the applications of the leading-edge suction theory, the pitching moment as a function of the theoretical lift data was calculated and plotted, shown in Figure 26. Equations (15) and (16) [35] were used to calculate the pitching moment for the leading-edge suction theory lift values. The pitching moment at zero lift, C_{m_0} used in equation (15), is the zero-lift pitching moment for the wind tunnel model (0.0039). Equation (15) includes the nonlinear pitching moment component as previously described

in section 5.2.1. To solve for the linear pitching moment component, equation(16), the lift curve slope of the leading-edge suction theory ($C_{L\alpha_{wLES}} = 0.0501$ per degree) was used. The fuselage and tail components in equation (16) are treated the same as they were in section 5.4, above.

$$C_m = C_{m_0} + C_{m\alpha_{linear}}\alpha + C_{m_{Nonlinear}}(\alpha) \quad (15)$$

$$C_{m\alpha_{linear}} = C_{L\alpha_{wLES}}\left(\frac{x_{cg}-x_{ac}}{\bar{c}}\right) + C_{m\alpha_f} - \eta V_H C_{L\alpha_t}\left(1 - \frac{d\varepsilon}{d\alpha}\right) \quad (16)$$

$$C_{m_{Nonlinear}}(\alpha) = C_{LVortexLift}(\alpha)\left(\frac{x_{cg}-x_{ac}}{\bar{c}}\right) \quad (17)$$

The method used to find the pitching moment as a function of the leading-edge vortex lift values has been found to not accurately predict the pitching moment of the wind tunnel model, Figure 26. The baseline pitching moment slope is -0.0280 and the leading-edge suction theory pitching moment slope is -0.0147. The error between theory and the baseline results is in a 47.5% error.

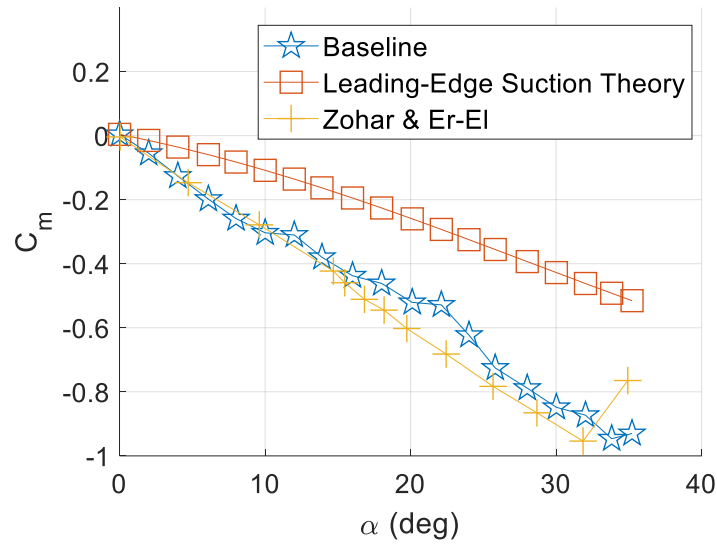


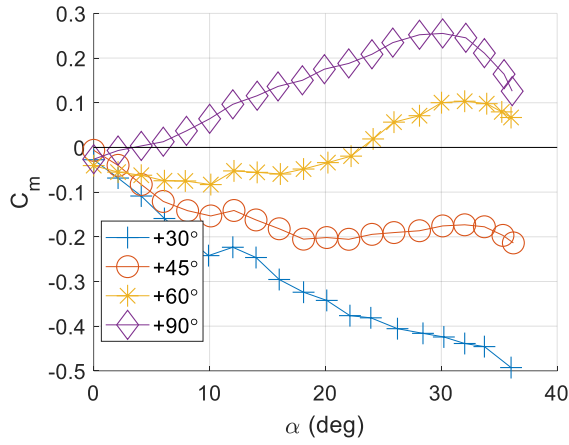
Figure 26. Pitching Moment Comparison between Wind Tunnel Data and Data from Zohar and Er-El [6, 8, 30]

5.4.2 Deflected Wing Tip Delta Wing

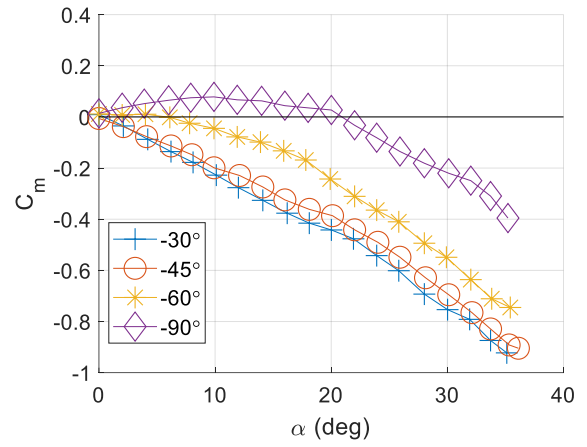
The significant result from the positive wing tip configuration tests is the relationship between deflection angle and static longitudinal stability. Figure 27 a) identifies that the model's pitching moment slope changes from negative to positive as wing tip deflection angle is increased. The change in pitching moment slope means that the model transitions from a statically stable configuration, a positive static margin, to a statically unstable configuration, a negative static margin. At the 90-degree positive wing tip deflection, the model is completely unstable with a pitching moment slope of 0.0094. Wing tip deflection angles of 60 and 45 degrees cause the model to be nearly neutrally

stable (static margin equal to 0), while the deflection angle of 30 degrees remains statically stable. An interesting trend observed for the 30 through 60-degree positive wing tip deflections is that between 10 and 15 degrees angle-of-attack, there is a jump in pitching moment. This could be due to the onset of the vortex lift component, as this is the expected angle-of-attack range for which the leading-edge vortex is fully formed [12].

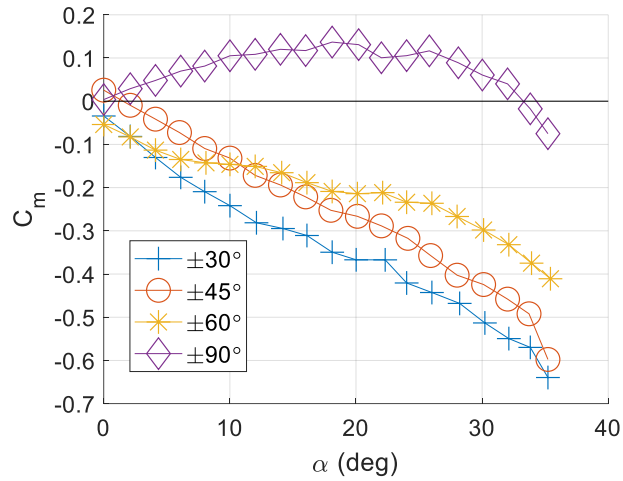
The relationship between an increase in wing tip deflection angle and a decrease in longitudinal stability that was identified for the positive wing tip deflection is present again for negative wing tip deflection, as seen in Figure 27 b). A significant result for the negative wing tip configuration is the pitching moment at a deflection of negative 90 degrees. At angles-of-attack from 0 to 20 degrees, the model is neutrally stable, and with further increase in angle-of-attack, the stability characteristics of the model become statically stable with a pitching moment slope of -0.0245. While the negative 90-degree wing tip layout experienced variations in its stability characteristics, the rest of the negative wing tip layouts were statically stable throughout the angle-of-attack range. The asymmetric configuration, Figure 27 c), shows that for wing tip deflections of 30 and 45 degrees, the model is statically stable. At the asymmetric wing tip deflection of 60-degrees, the model is predominantly neutrally stable for angles-of-attack between 4 and 22 degrees. At higher angles-of-attack, the 60-degree configuration is longitudinally stable. The 90-degree asymmetric wing tip deflection is largely neutrally stable for all angles of attack.



a) Positive Wing Tip Deflection



b) Negative Wing Tip Deflection



c) Asymmetric Wing Tip Deflection

Figure 27. Various Wing Tip Deflection Pitching Moment Plots

5.4.3 Comparison between Configurations

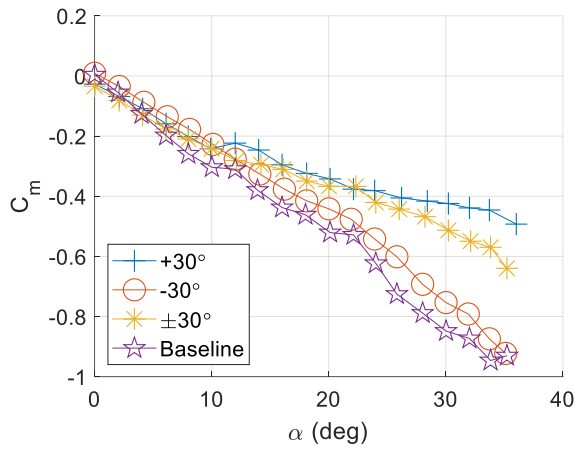
The most significant trend observed for the various wing tip configurations is the change in static longitudinal stability with wing tip deflection. Table 5 represents the most drastic examples of stability change for the three 90-degree wing tip deflection layouts compared to the baseline.

Table 5. Summary of Longitudinal Stability Characteristics of the Positive and Negative 90-Degree Wing Tip Deflections

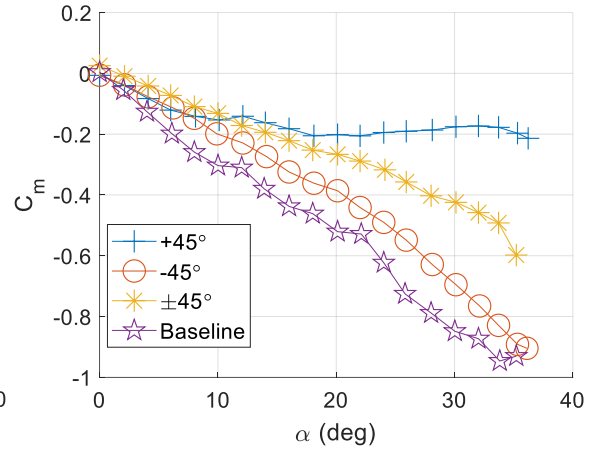
	$C_{m\alpha}$		
	Stable	Neutrally Stable	Unstable
Positive	-	-	0.0094
Asymmetric	-	~ 0	-
Negative (Pitch)	-0.0245 (20-34°)	0 (0-20°)	-
Baseline	-0.0280	-	-

As the wing tips are deflected from 0 to 90 degrees in the positive direction, the pitching moment slope changes from a stable configuration, at the baseline, to an unstable configuration as listed in Table 5. Unlike the positive wing tip configuration, the negative deflection does not show as drastic of a change in pitching moment slope with an increase in wing tip deflection. Interestingly, the negative wing tip layout has a very similar pitching moment slope to the baseline during its stable phase. Once the wing tips are completely deflected down to negative 90 degrees, the model is neutrally stable. The longitudinal

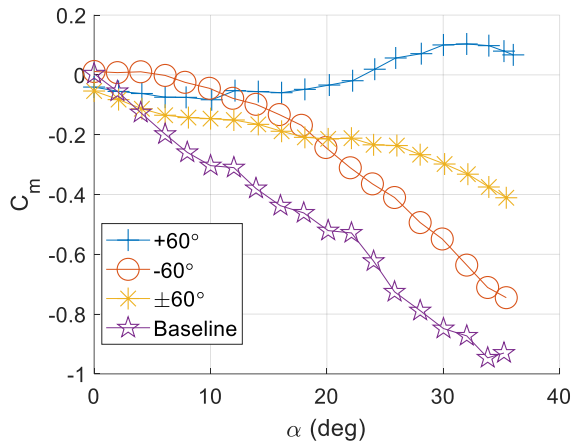
stability shift, observed for all the wing tip configurations, implies that the wing tips are interacting with the flow over the wing in such a way that shifts the center of lift forward. Figure 28, below, presents each wing tip deflection grouped by its absolute deflection value while also in reference to the statically stable baseline. Here it is observed that the positive wing tip deflection has a more unstable pitching moment slope, while the negative has the most stable slope, and the asymmetric layout falls in between the previous two.



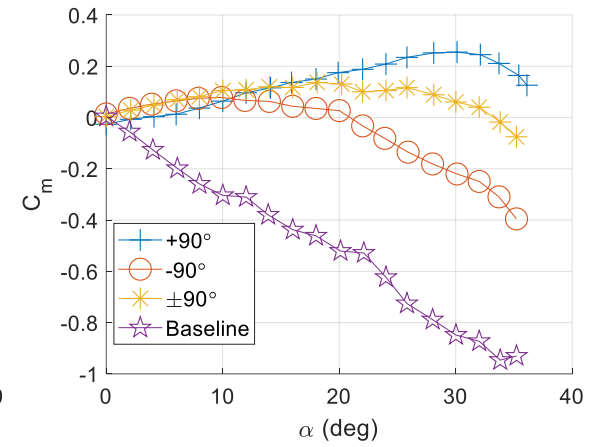
a) 30° Wing Tip Deflection



b) 45° Wing Tip Deflection



c) 60° Wing Tip Deflection



d) 90° Wing Tip Deflection

Figure 28. Comparison of Pitching Moment Data Across Each Wing Tip Deflection

Configuration

5.5 Lateral-Directional Tests

For an aircraft to have static lateral stability, the change in rolling moment with sideslip angle needs to be negative, Figure 29 a). An aircraft that is laterally stable will produce a restoring roll response that results in level wings when subjected to a roll disturbance. An aircraft has a statically stable yaw moment when the change in yaw moment with sideslip angle is positive, Figure 29 b). Directional stability is also known as weathercock. A directionally stable aircraft will always point into the relative wind direction when subjected to a yaw disturbance [35].

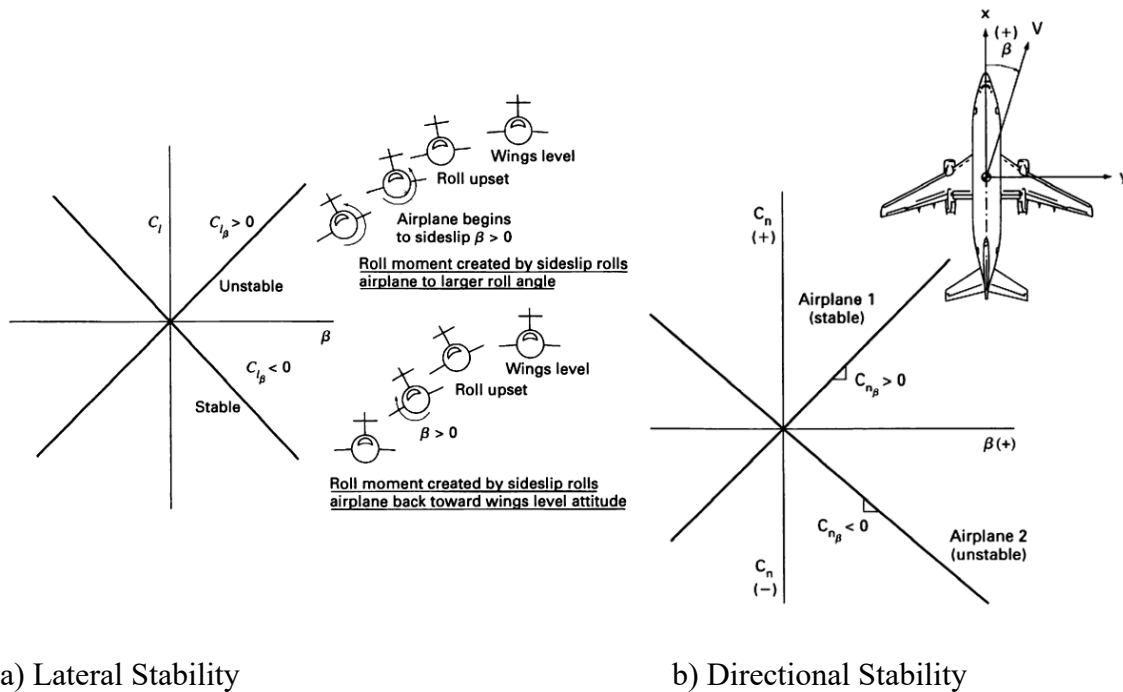


Figure 29. Static Lateral-Directional Stability [35]

5.5.1 Baseline Delta Wing in Sideslip

Figure 30 shows the lateral (a) and directional (b) stability of the baseline wing. The baseline model has a positive rolling slope from 0-6 degrees angle-of-attack and then a negative rolling slope for every angle-of-attack after 6 degrees. This means that the aircraft is laterally unstable for low angles-of-attack but stable for high angles-of-attack. The aircraft has a positive yaw moment slope which means it is directionally stable. In each of the plots shown below, the rolling and yawing moment share a plateau region between 10-15 degrees of sideslip. The phenomena driving the plateau is unclear but may be due to a shedding leading-edge vortex interacting with the vertical tail.

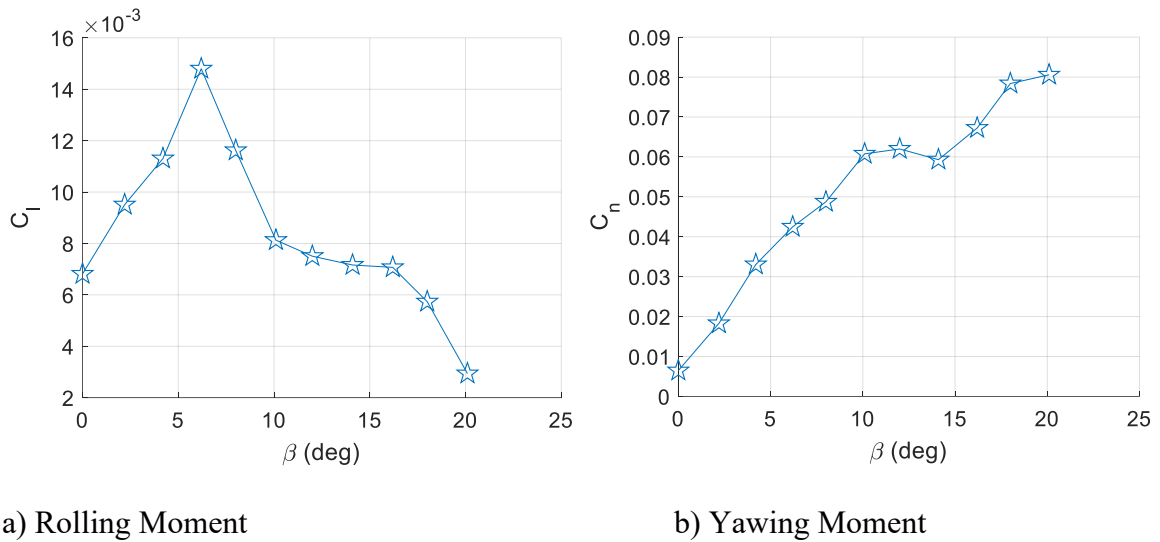
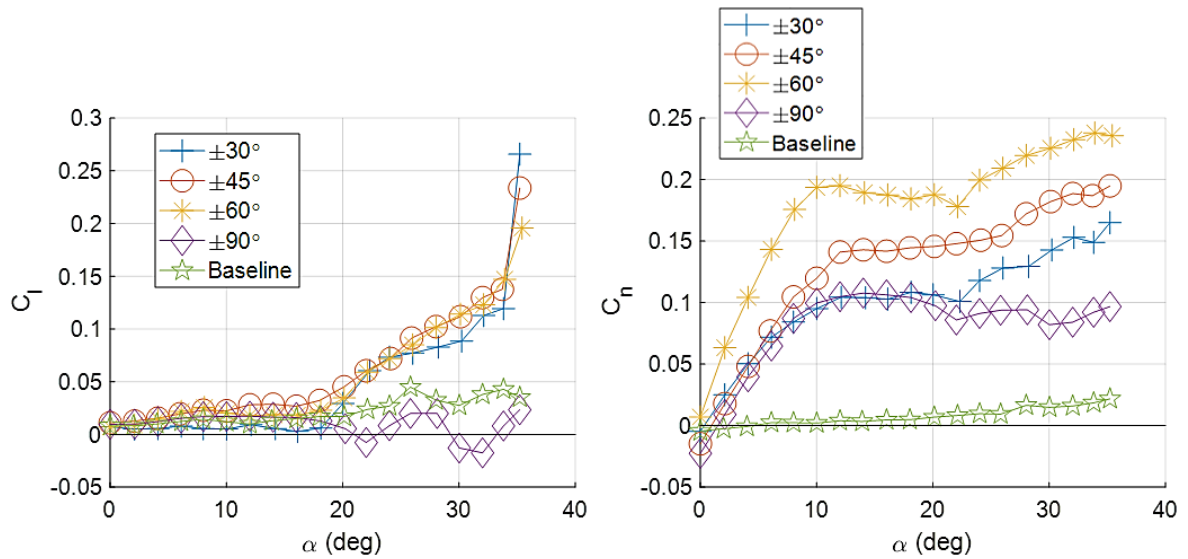


Figure 30. Baseline Lateral and Directional Stability vs Sideslip

5.5.2 Delta Wing Configurations in Pitch

The asymmetric 30, 45 and 60-degree layout show a strong and steady increase in rolling moment due to the deflected tips acting as partial ailerons, Figure 31 a). The rolling moment instability of the baseline and asymmetric 90-degree layout at high angles of attack could be attributed to wing rock vortex bursting. Wing rock vortex bursting at high angles of attack near stall has been documented for similar delta wings [4, 11]. In Figure 31 b), the yawing moment coefficients increase significantly due to deflected wing tips in comparison to the baseline layout. Here the yawing moment increases up until wing tip deflections of 60 degrees, then the 90-degree deflection falls between the baseline and 30-degree deflection. The baseline layout showed virtually no change in yawing moment at low angles-of-attack, which is expected due to aircraft symmetry. The slight increase in the baseline yawing moment at high angles of attack can be attributed to vortex bursting as the stall limit approaches. At a relatively stable 16 degrees angle-of-attack, the yawing moment coefficient for the 30, 45, 60 and 90-degree asymmetric wing tip layout are: 0.108, 0.145, 0.184 and 0.104, respectively. These values indicate that the asymmetric wing tip deflection is effectively increasing vertical tail area and in turn increasing the yawing moment.



a) Rolling Moment

b) Yawing Moment

Figure 31. Asymmetric Wing Tip Deflection Lateral and Directional Stability vs Pitch

5.6 Reynolds Number Effects

To understand the effect Reynolds number has on the aircraft model and the various delta wing layouts, two Reynolds number tests were ran on three different configurations. The Reynolds numbers were 3.65×10^5 and 5×10^5 . The three configurations were: (1) baseline, (2) positive 45-degree wing tip deflection, and (3) positive 90-degree wing tip deflection. Figure 32 and Figure 33, below, show that the Reynolds number does not impact the lift curve slope or pitching moment of the model for all three delta wing configurations. The lack of a Reynolds number impact on the aerodynamic characteristics of the model under test could be due to the relatively low order and closeness of the values used in the

tests. As detailed in section 1.1.3, an increase in Reynolds number should change the vortex characteristics. This relationship was observed for a range of Reynolds numbers between 6 and 60 million, which corresponds to one to two magnitudes larger than the Reynolds numbers used in the experiments for this thesis. Henderson's observation on the relationship between experimental and potential flow lift with increasing Reynolds number [16] would result in different lift curve slopes with increased Reynolds numbers. This effect is not observed for the data presented here. The expected impact of Reynolds number on the aerodynamic characteristics of a delta wing may not be evident until larger order Reynolds numbers are tested.

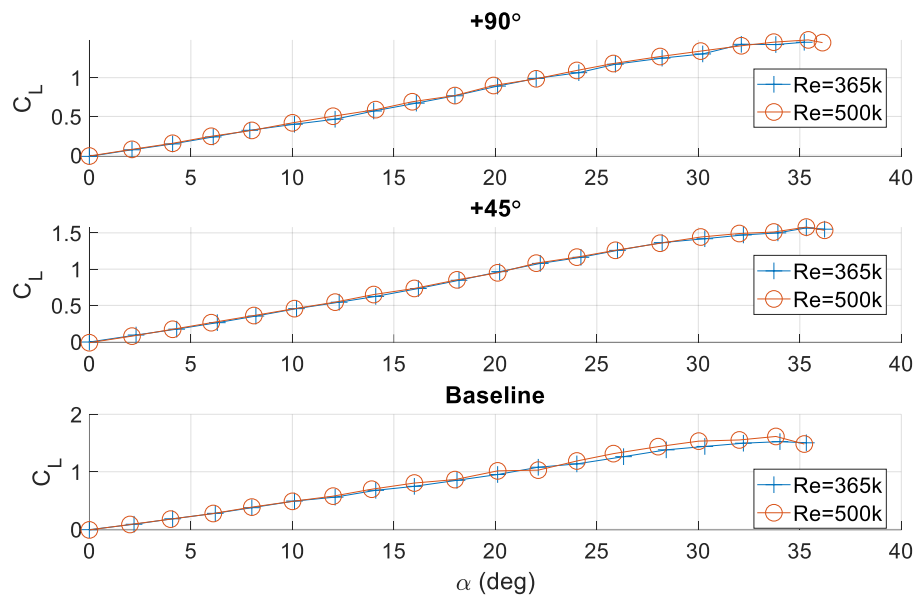


Figure 32. Lift Curve Slope with Varying Reynolds Number

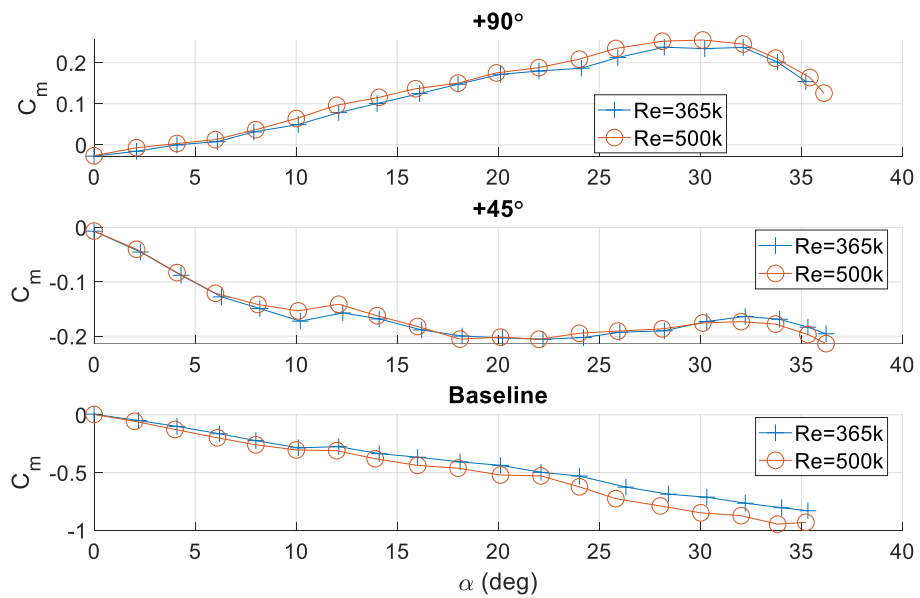
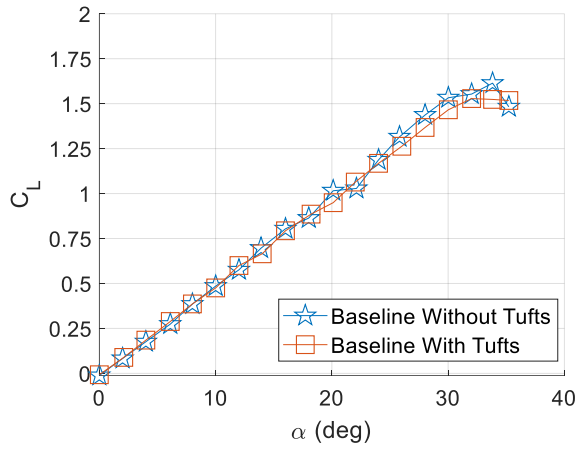


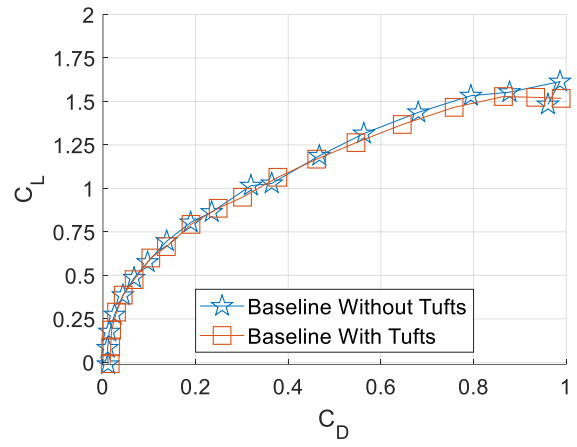
Figure 33. Pitching Moment with Varying Reynolds Number

5.7 Flow Visualization

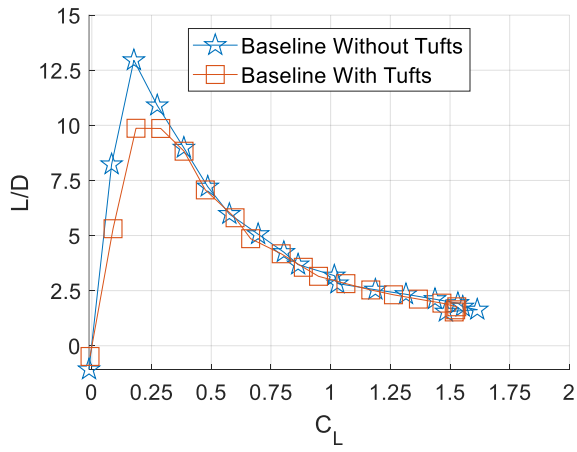
An aerodynamic loads study was performed while the tufts were installed to observe how they affected the aerodynamic characteristics of the model and to understand if the flow characteristics pictured in this section represent the flow field of the data presented previously. Figure 34 a) & b) shows that the lift curve slope and drag polar do not significantly change with the addition of tufts. On the other hand, the aerodynamic efficiency and pitching moment have changed. The maximum lift-to-drag ratio of the baseline with and without tufts are, 9.87 and 12.95, respectively. The maximum lift-to-drag ratio decreases due to the added skin friction drag of the tape and tufts and the minor change in lift characteristics. The pitching moment slope of the baseline with and without tufts are, -0.0393 and -0.0280 per degree. These results show that the aerodynamic efficiency and pitching moment are more sensitive to wing surface roughness.



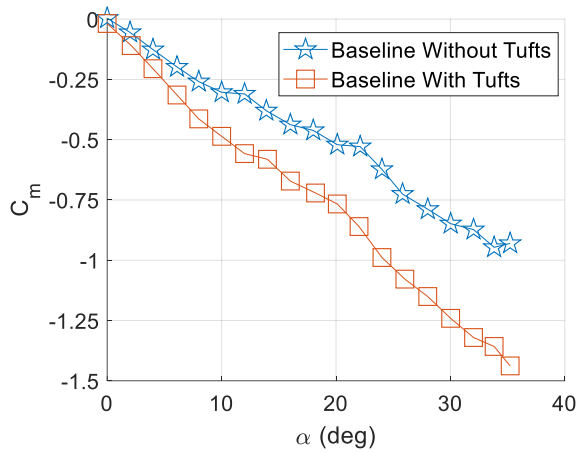
a) Lift Curve Slope



b) Drag Polar



c) Aerodynamic Efficiency



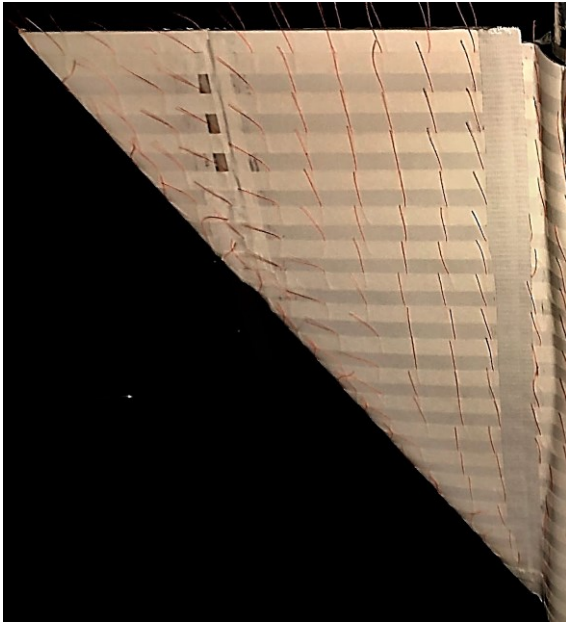
d) Pitching Moment

Figure 34. Baseline Aerodynamic Characteristics with and without Flow Visualization

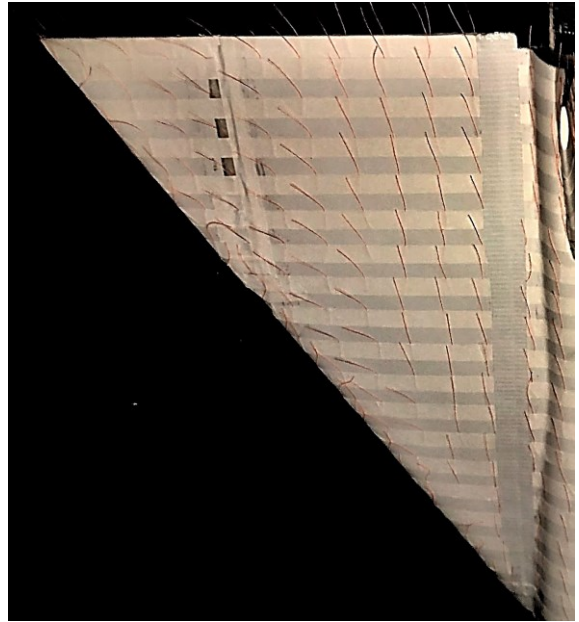
Tufts

Figure 35 displays three flow visualization images using tufts at three angles-of-attack. The direction of flow in the images below, is from the bottom of the image to the top and parallel to the fuselage. Moving from right to left, the images increase with angle-of-attack. The leading-edge vortex is also seen in each image and is represented by the orange tufts slanted towards the leading-edge. The leading-edge vortex is observed to increase in size as angle-of-attack is increased. The increase in vortex size is evident by the increasing number of tufts that are oriented towards the leading-edge and the decrease in tufts that are in line with the flow direction.

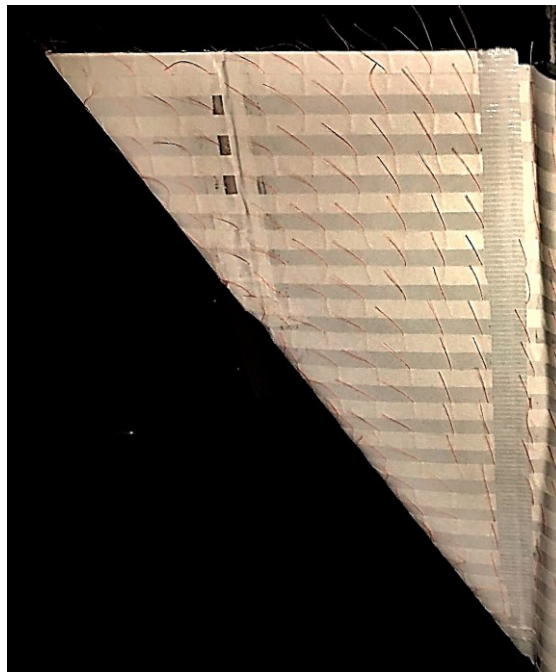
In Figure 35 a), the leading-edge vortex is concentrated near the wing's leading-edge, whereas in Figure 35 c) it has increased in size enough to engulf the whole wing. The crossing of the tufts just inboard of the leading-edge is due to the small stretch of attached flow that occurs before the flow separates into a vortex. This effect is distinct to a rounded leading-edge delta, as detailed in section 1.1.2 above. The rounded leading-edge is a key design characteristic of the delta wing used in this study. For more flow visualization images, refer to Appendix A. Flow Visualization Images.



a) $12^\circ \alpha$



b) $16^\circ \alpha$



c) $20^\circ \alpha$

Figure 35. Half Wing Flow Visualization Using Tufts

CHAPTER 6. SUMMARY AND CONCLUSIONS

The flow over a delta wing is unique due to the vortex element that separates off of the leading-edge of the wing. The vortex provides additional lift generation and enables a delta wing to produce lift for much higher angles-of-attack than that of a rectangular wing. These unique flow characteristics of a delta wing are the focus of this thesis. The low-speed aerodynamic characteristics of a novel delta wing layout with deflected wing tips have been investigated and presented. The primary motivation to investigate a delta wing and its subsonic capabilities was to advance the research and development of UAV technologies that are of current interest at The Ohio State University Aerospace Research Center.

The delta wing of interest in this study had a leading-edge sweep of 60 degrees, a high-speed airfoil with a rounded leading-edge, and novel wing tips that can rotate a full 180 degrees about a hinge in the wing that is parallel to the fuselage axis. Three different wing tip configurations were tested at a Reynolds numbers of 3.65×10^5 and 5×10^5 , based off the mean aerodynamic chord. The three configurations were: positive wing tip deflection with the wing tips deflected upward, negative wing tip deflection with the tips deflected downward, and asymmetric wing tip deflection. The asymmetric deflection is when one wing tip is deflected down, and the other wing tip is deflected up. Two methods were used to assess the aerodynamic characteristics of the aircraft model, a panel method code, and wind tunnel model testing.

The panel method code used in this study was OpenVSP. It was found that the panel method code accurately predicted the lift characteristics of public delta wing data, delta wing lifting theory, and data gathered from the baseline layout, but only after implementing a vortex lift factor. This additional factor was expected because panel method codes only solve the potential flow model, which does not account for vortex lift. Additionally, the adjusted panel method results did not accurately predict the pitching moment characteristics detailed in public delta wing data or the data gathered for the baseline delta wing layout. The panel method code OpenVSP, was found to be useful for initial design analysis but may not be appropriate for detailed flight vehicle design and analysis.

Wind tunnel testing was performed at ARC's 3'x5' subsonic wind tunnel. Aerodynamic data was recorded using a six-component internal force balance. The baseline configuration's aerodynamic characteristics were initially compared to outside public data and theoretical expectations in order to validate the wind tunnel model. It was found that the baseline delta wing configuration used in this study closely agreed with public data and delta wing lift theory. Upon comparing the different Reynolds number tests, it was found that changing the Reynolds number from 3.65×10^5 to 5×10^5 showed minimal change in the aerodynamic characteristics. Significant trends identified from the wind tunnel results show that by rotating the wing tips, in either the positive or negative direction, the longitudinal stability characteristics of the model change drastically without substantial loss in aerodynamic lift capability. These findings introduce the capability for potential real-time enhancements in control and maneuverability flight operations.

6.1 Future Work

The capabilities of the deflected wing tips have been extensively researched for variations in angle-of-attack. A next step in assessing the novel delta wing platform would be to run all the deflected wing tip configurations in side-slip tests. Side-slip tests for all of the various wing tip deflections would provide insight into how the wing tips impact the lateral and directional stability of the aircraft model. Another set of tests include running the model at various angles-of-attack, while sweeping through side-slip angles. For the coupled pitch and side-slip tests, a new sting mounting bracket to the parallelogram linkage needs to be designed and manufactured. To fully understand the stability characteristics of the model, control surfaces need to be installed on the model to assess the elevator and rudder effectiveness for all the wing tip configurations.

Other delta wing planforms can also be easily assessed going forward as a result of the modular design of the fuselage. Potential delta wing layouts include a double delta and an ogive delta. These new planforms could also be modified with rotating wing tips. It would be very interesting to see if the unique trends discovered in this thesis, on a standard delta wing, can also be detected on other delta wing types.

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Appendix A. Flow Visualization Images

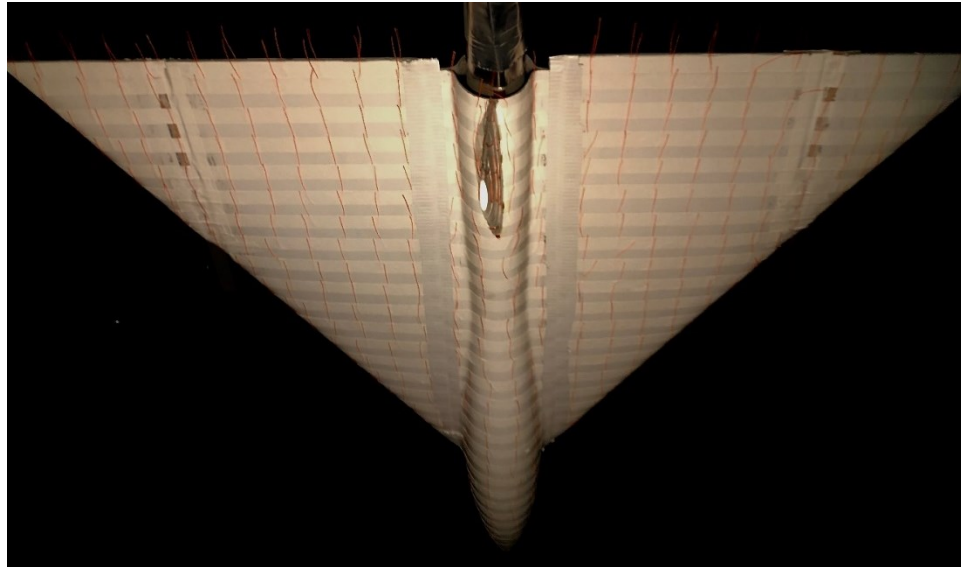


Figure 36. Top View – 0° Angle-of-Attack

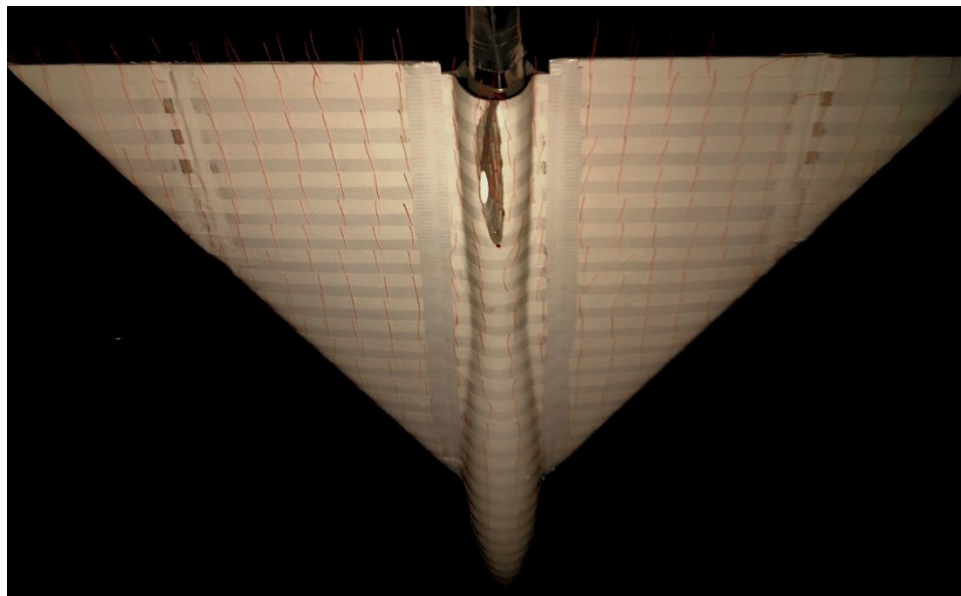


Figure 37. Top View – 4° Angle-of-Attack

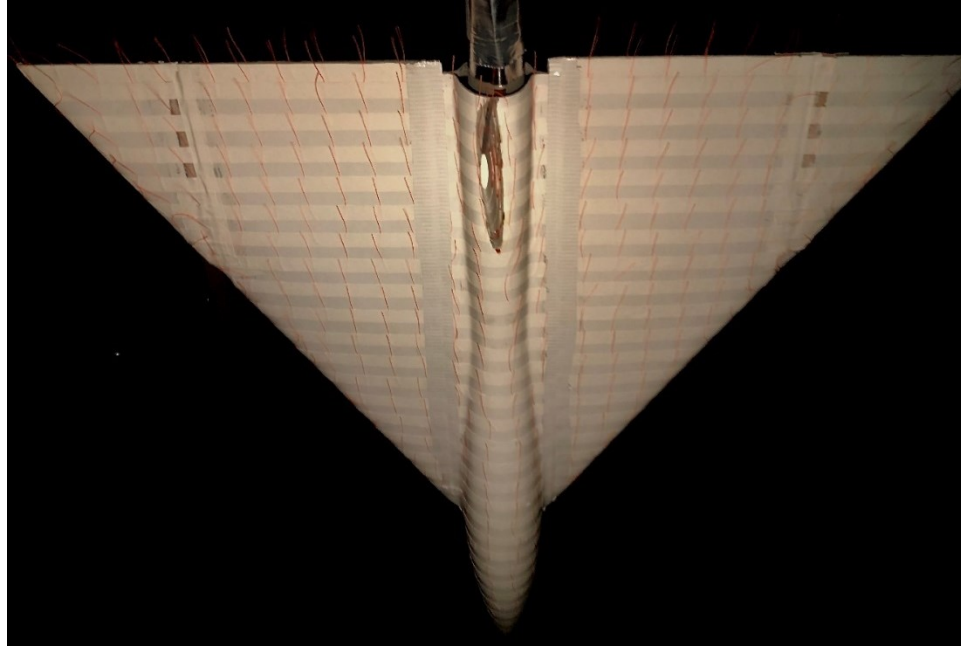


Figure 38. Top View – 8° Angle-of-Attack

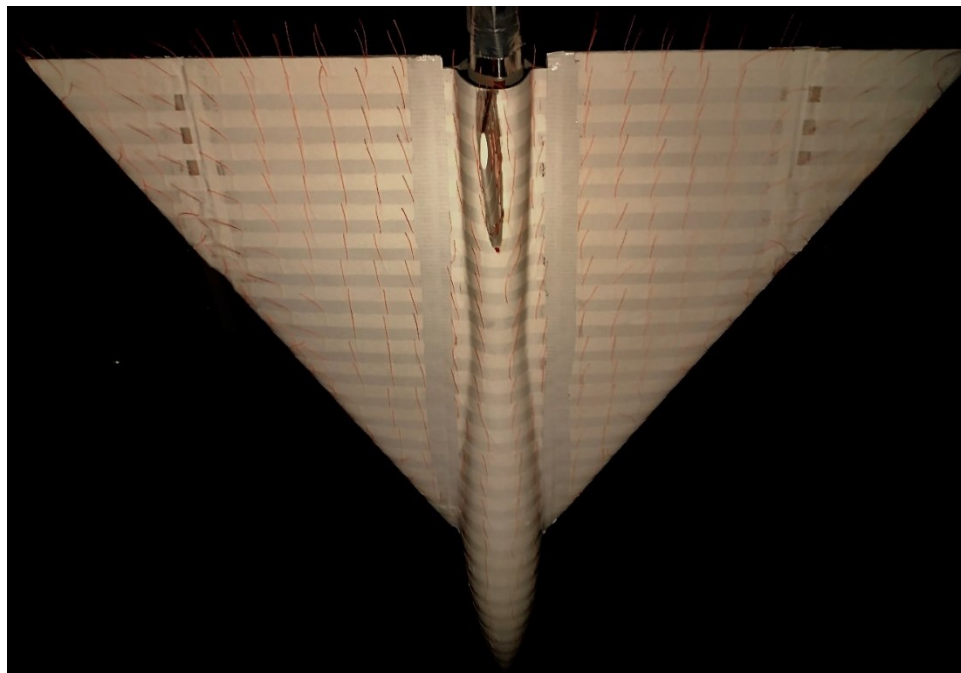


Figure 39. Top View – 12° Angle-of-Attack

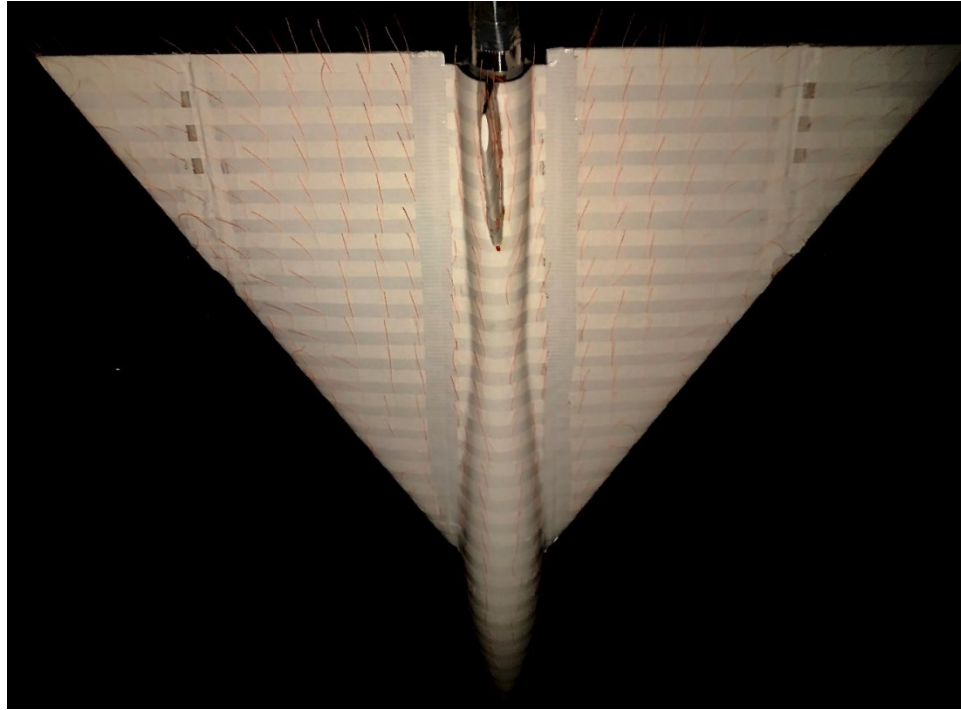


Figure 40. Top View – 16° Angle-of-Attack

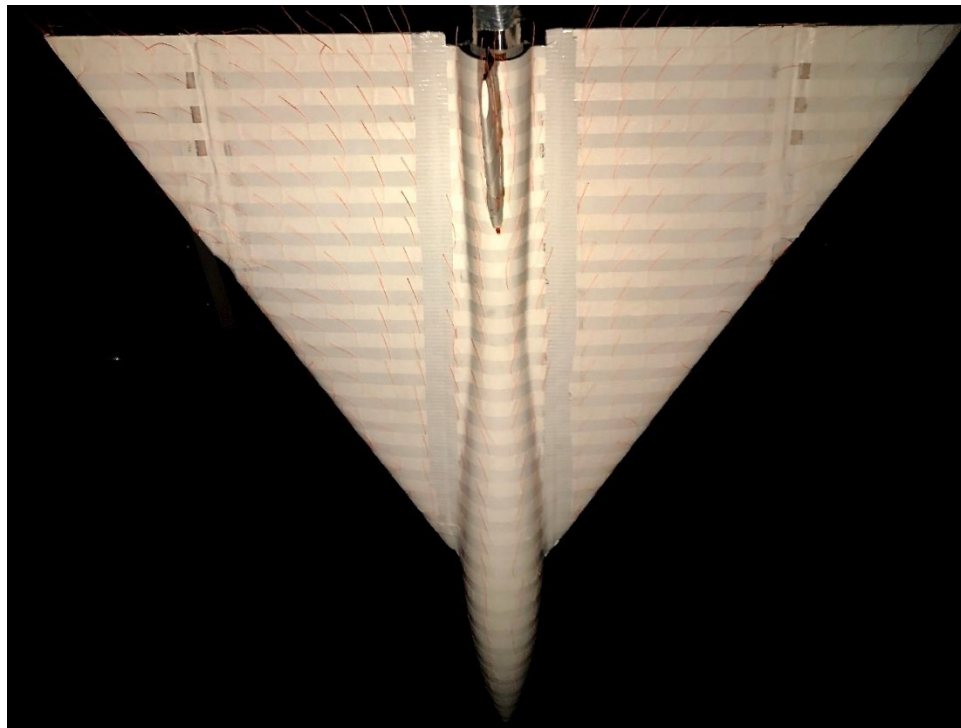


Figure 41. Top View – 20° Angle-of-Attack

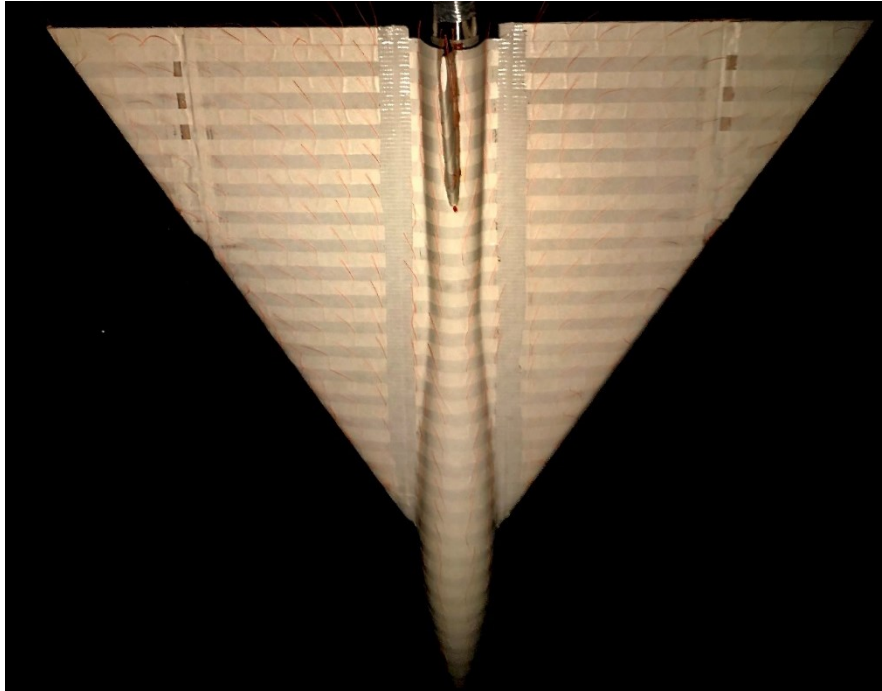


Figure 42. Top View – 24° Angle-of-Attack

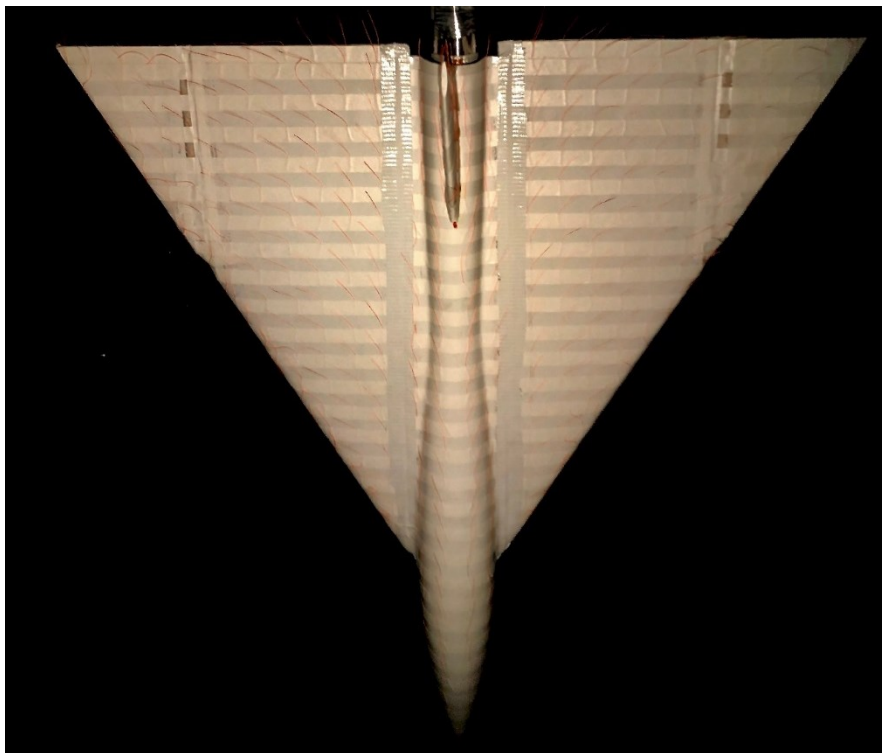


Figure 43. Top View – 28° Angle-of-Attack

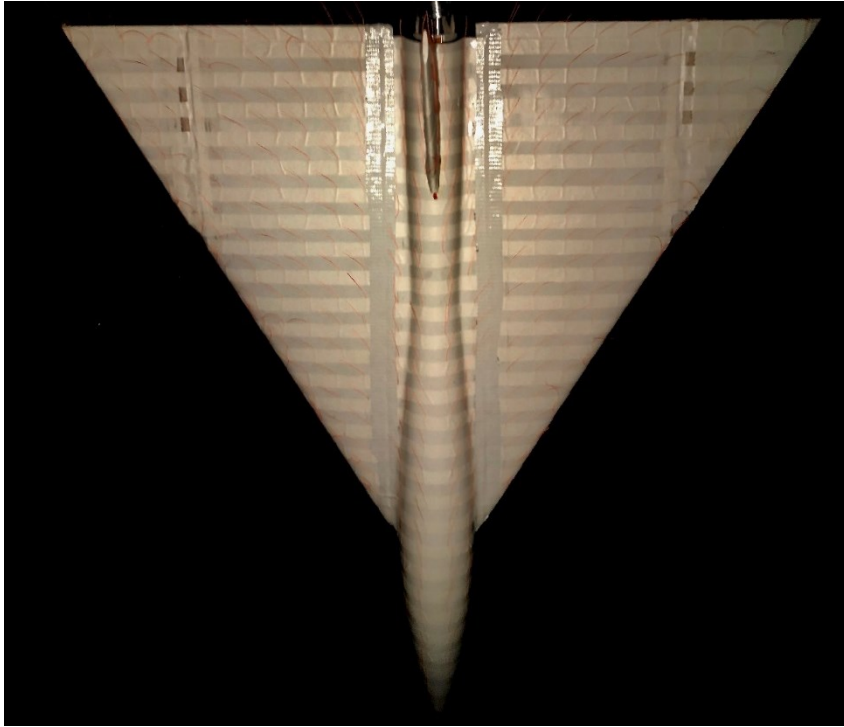


Figure 44. Top View – 32° Angle-of-Attack

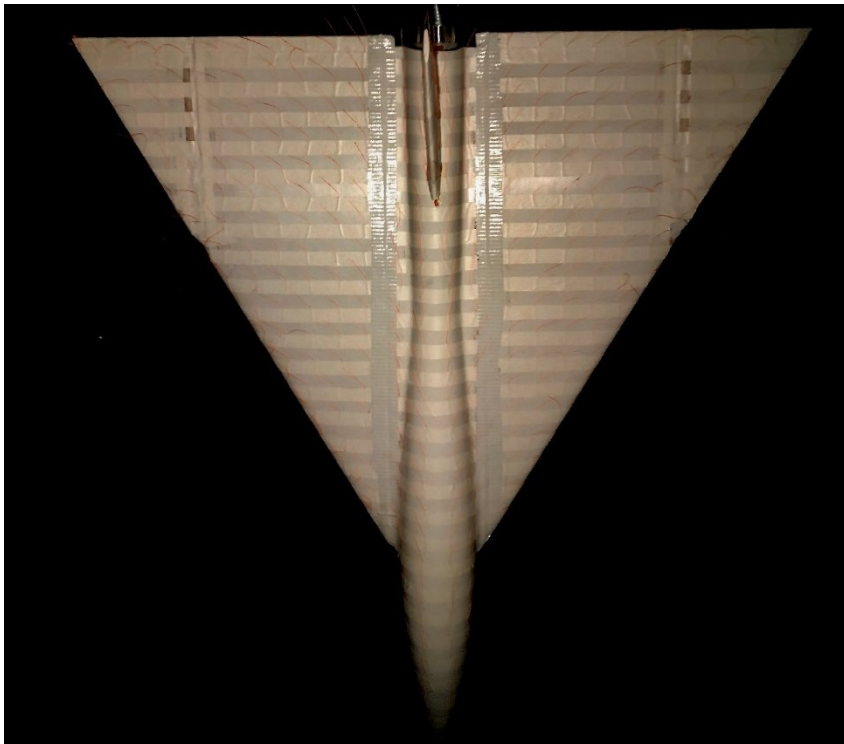


Figure 45. Top View – 36° Angle-of-Attack

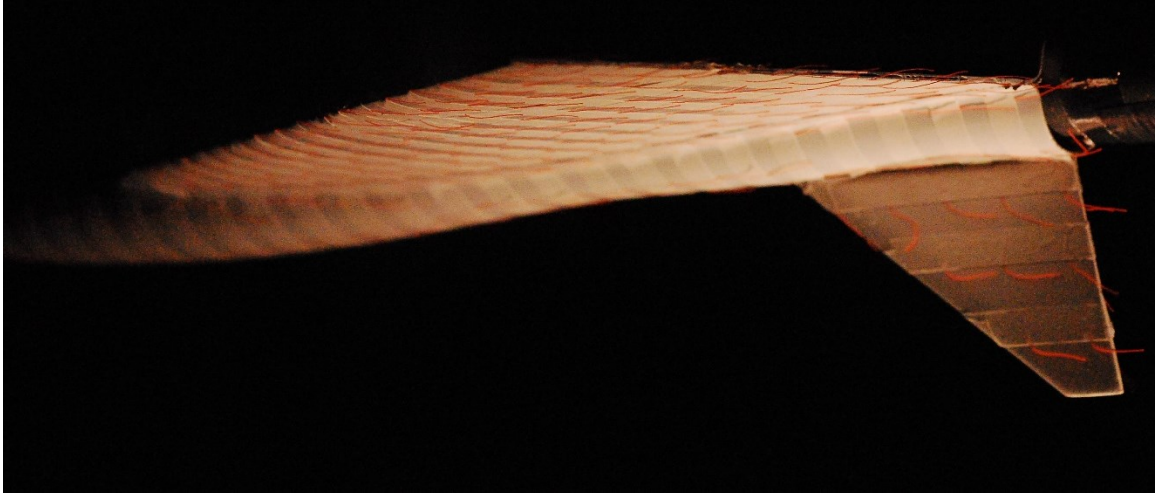


Figure 46. Side View – 0° Angle-of-Attack

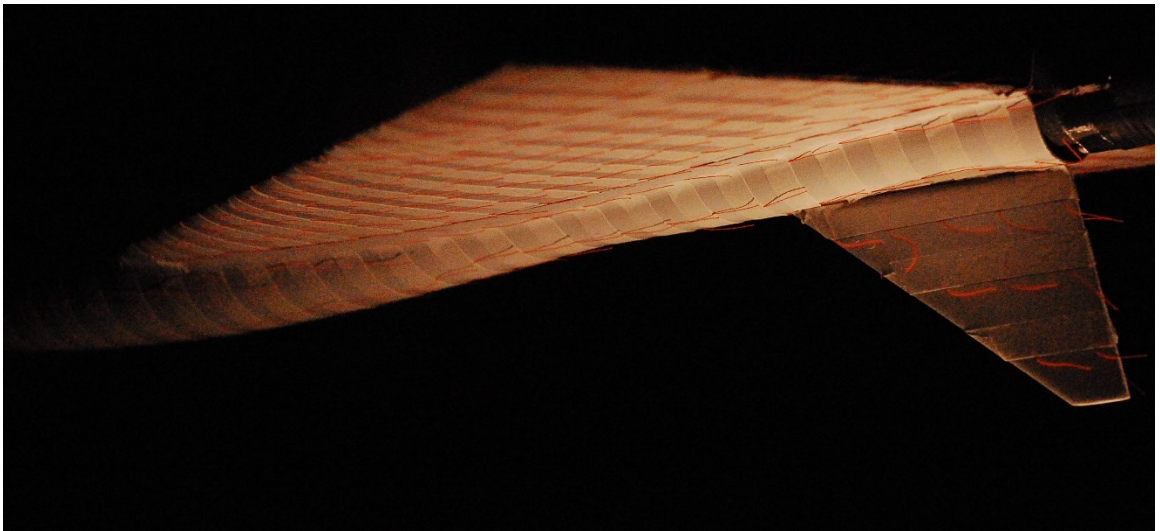


Figure 47. Side View – 4° Angle-of-Attack

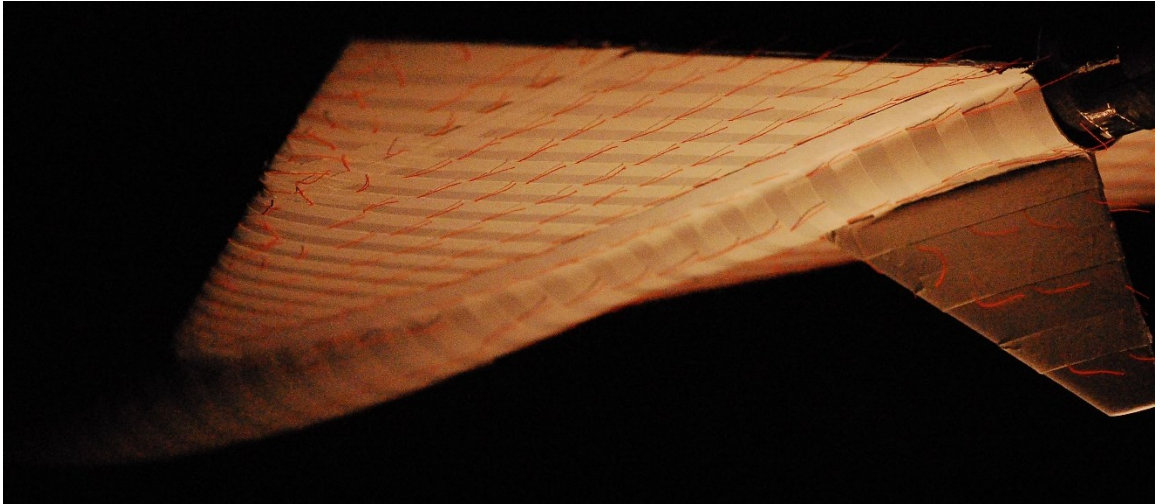


Figure 48. Side View – 8° Angle-of-Attack

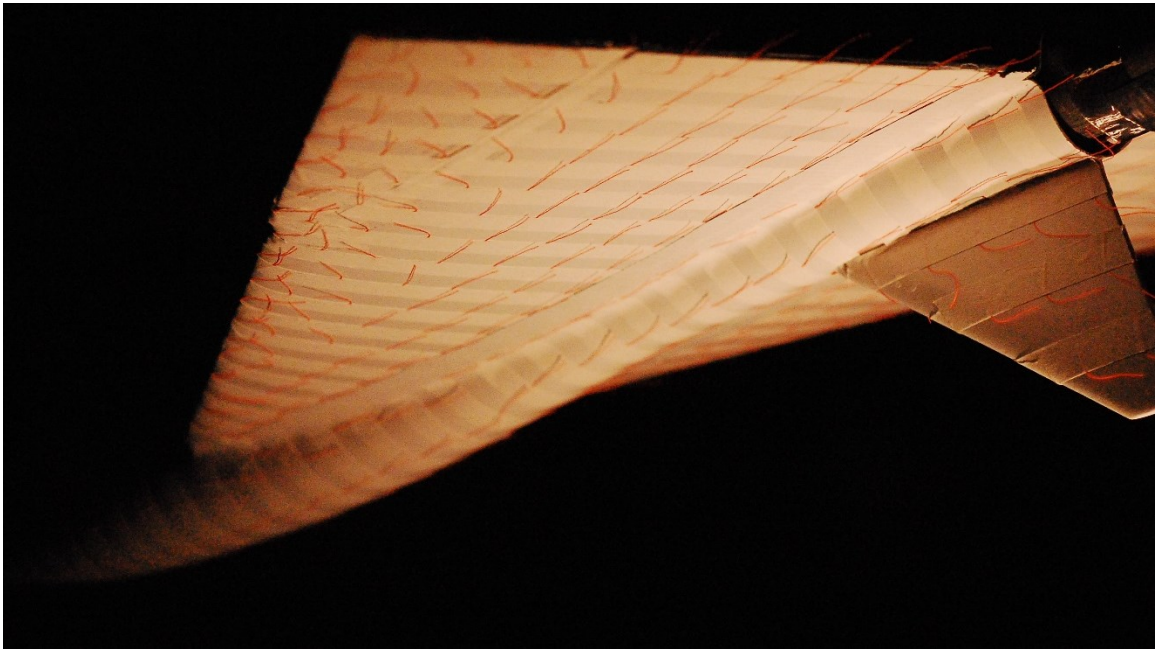


Figure 49. Side View – 12° Angle-of-Attack

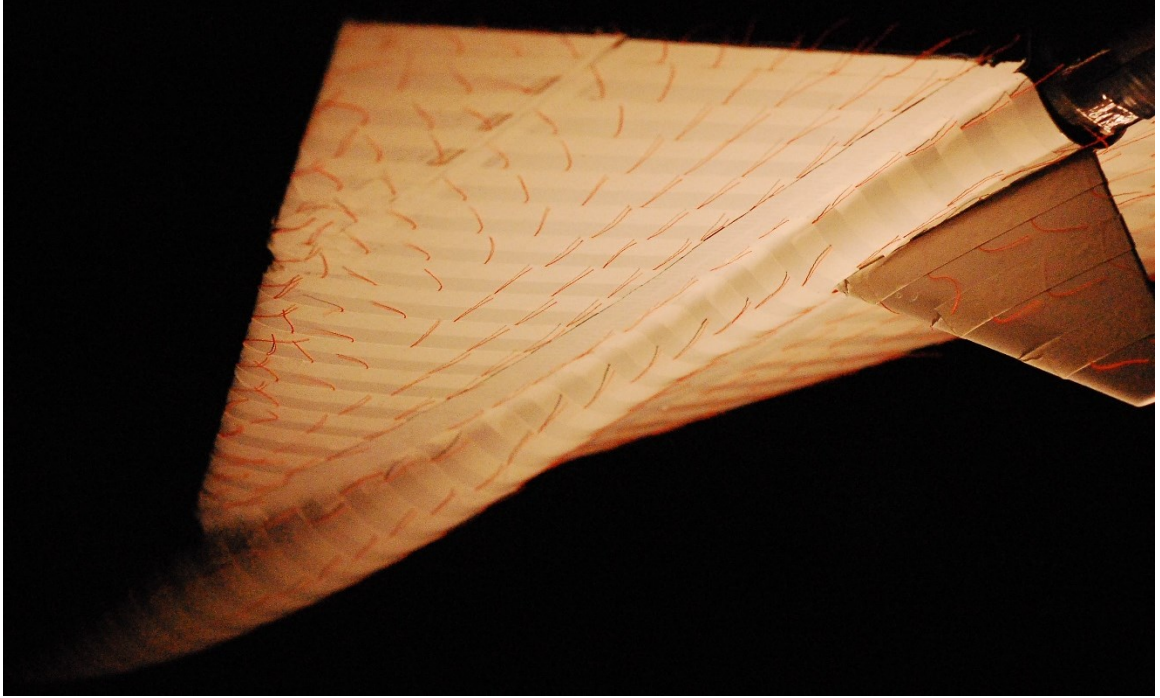


Figure 50. Side View – 16° Angle-of-Attack

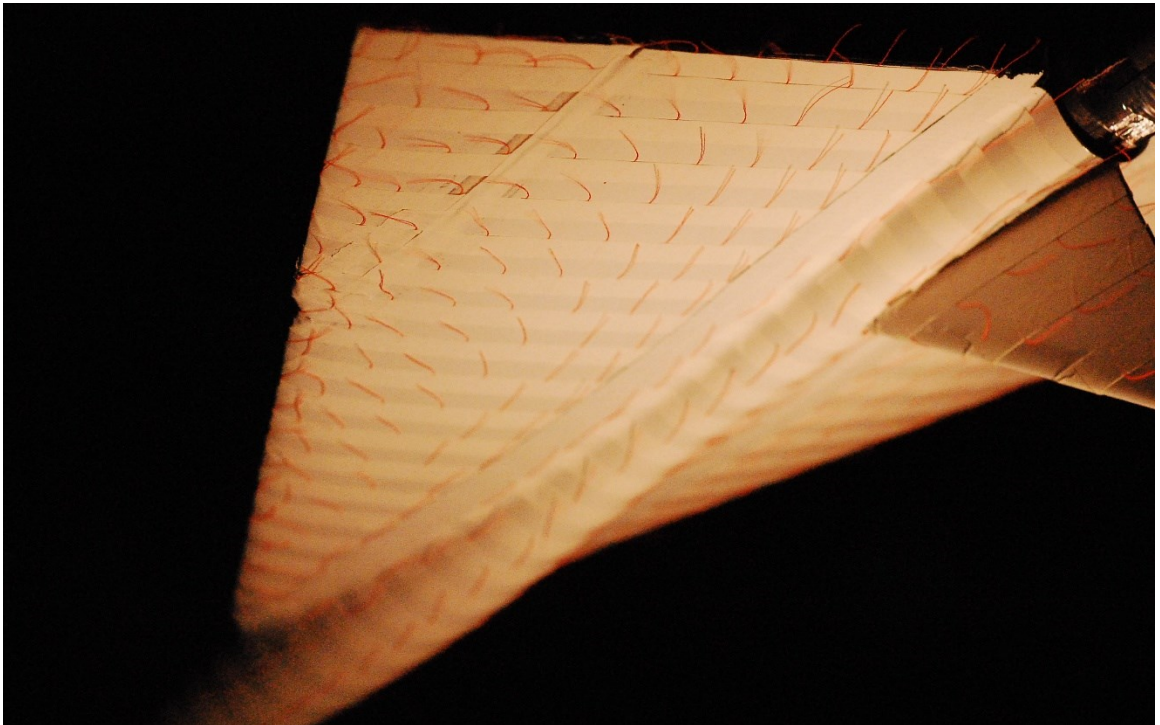


Figure 51. Side View – 20° Angle-of-Attack

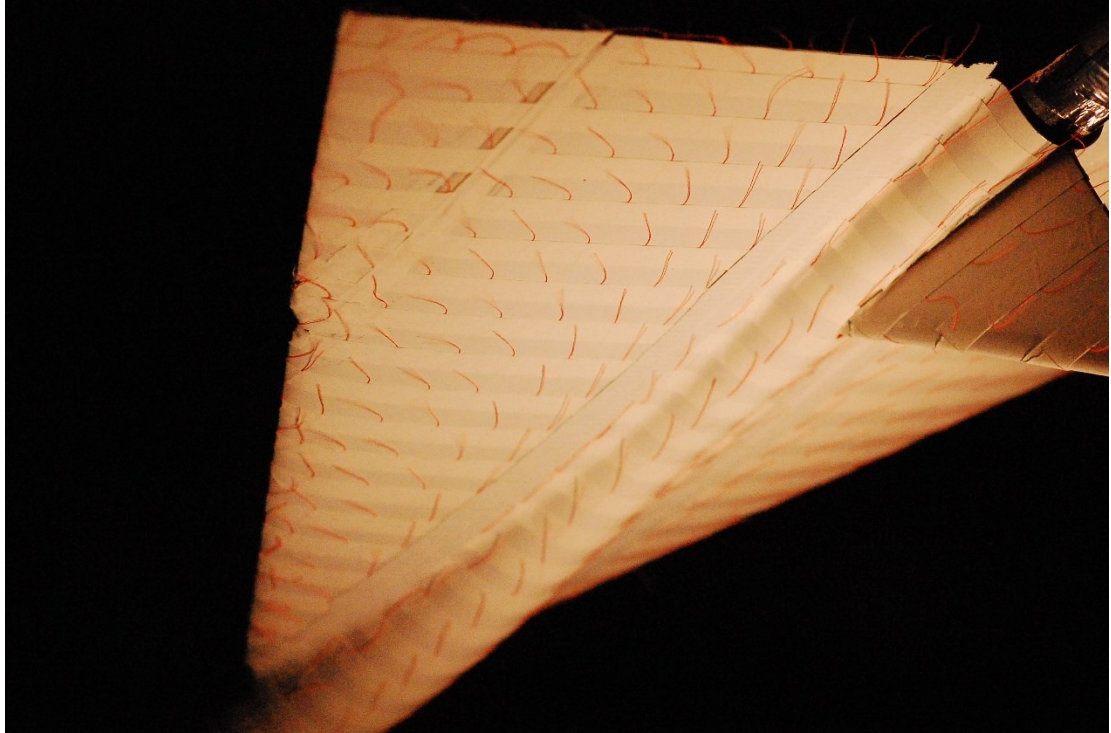


Figure 52. Side View – 24° Angle-of-Attack

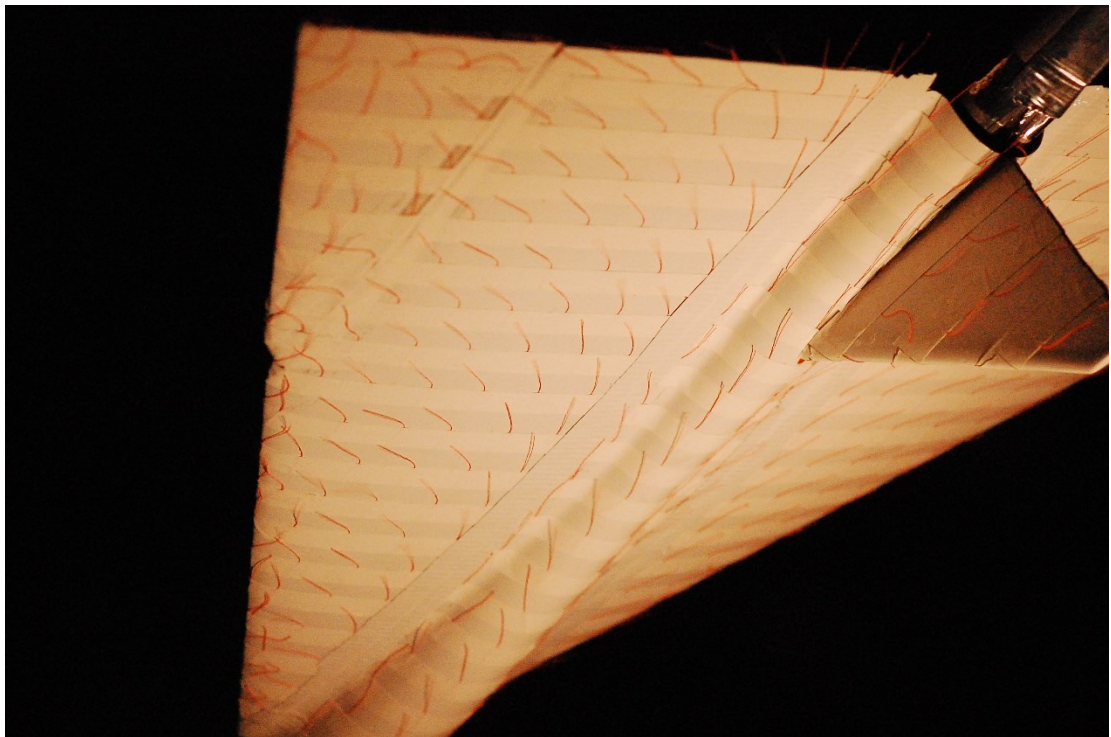


Figure 53. Side View – 28° Angle-of-Attack

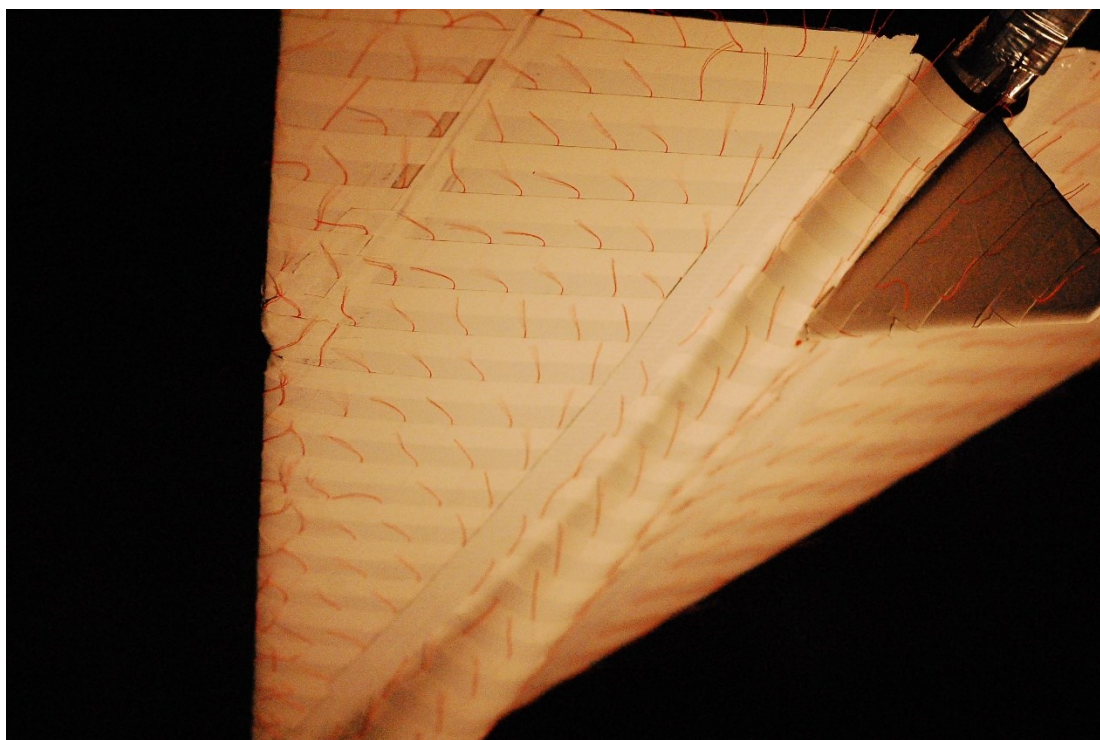


Figure 54. Side View – 32° Angle-of-Attack

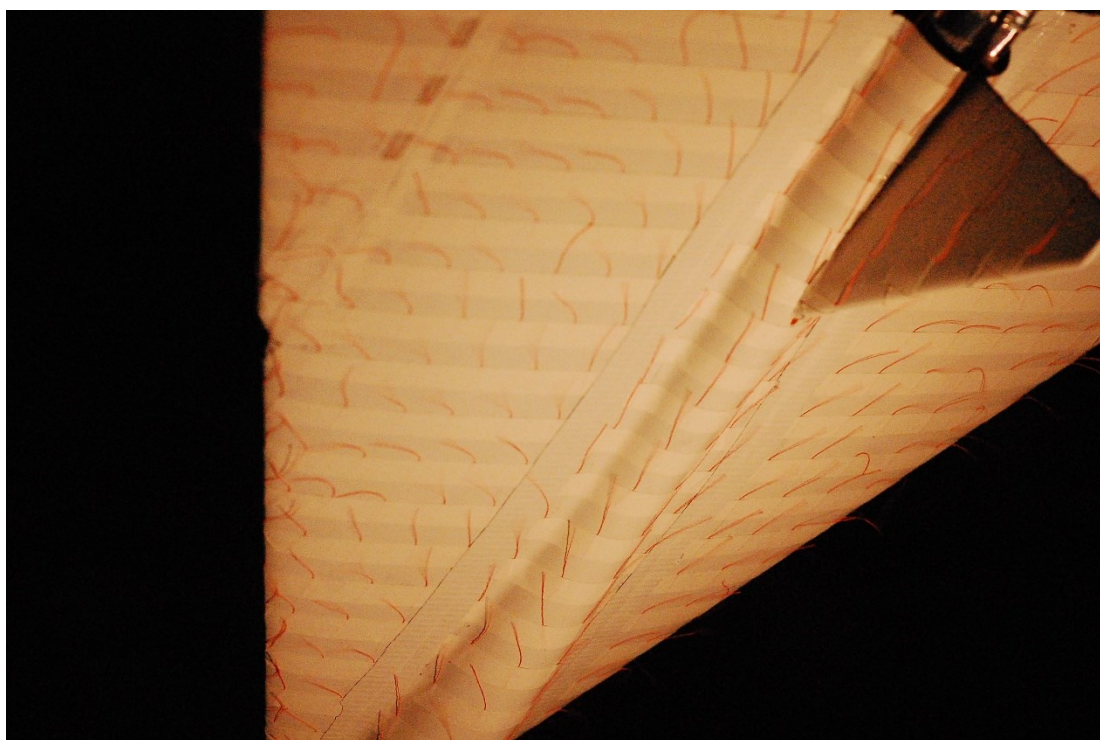


Figure 55. Side View – 36° Angle-of-Attack

Appendix B. Test Run Matrix

	Test #	Re x1000	Wing tip Deflection	Data Used in Thesis (Y/N)	Notes
Balance Data – Angle of Attack Sweeps	1	500	neg 90	N	Incorrect drag polar
	2	500	0	N	Incorrect drag polar
	3	365	0	Y	Fixed drag polar issue in 1-2
	4	500	0	Y	
	5	365	neg 90	N	
	6	500	neg 90	Y	
	7	365	pos 90	Y	
	8	500	pos 90	Y	
	9	365	pos 45	Y	
	10	500	pos 45	Y	
	11	365	neg 45	N	
	12	500	neg 45	Y	
	13	365	pos/neg 45	N	
	14	500	pos/neg 45	Y	Light gusting around 20-26 AoA
	15	500	pos/neg 90	Y	Light gusting around 8-12 AoA
	16	500	neg 30	Y	Windy
	17	500	pos 30	Y	Extra AoA at 36 for stall
	18	500	pos 60	Y	Extra AoA at 36 for stall
	19	500	neg 60	Y	Windy
	25	500	pos/neg 30	N	High winds closing tunnel doors - test ended early
	26	500	pos/neg 30	Y	Windy
	27	500	pos/neg 60	Y	Windy
Tufts	20	365	0		Lens tests
	21	365	0		Lens tests
	22	365	0		Lens tests
	23	365	0		Final images
Tufts and Balance Data	24	500	0	Y	
Balance Data - Side-Slip Sweeps	28	500	0	N	Model issues
	29	500	0	N	Model issues
	30	500	0	N	Model issues
	31	500	0	Y	Fixed issues from 28-30